DESIGN AND ANALYSIS OF LOW POWER, HIGH SPEED LEAKAGE-CONTROLLED DYNAMIC COMPARATOR FOR ADC APPLICATIONS USING 45NM CMOS TECHNOLOGY

ABSTRACT

Power dissipation has become one of the most critical design constraints in modern VLSI circuits, especially with aggressive CMOS technology scaling. As feature sizes shrink to deep sub-micron dimensions, sub-threshold leakage current increases exponentially due to threshold voltage reduction, becoming a dominant component of total power consumption in advanced technology nodes like 45nm CMOS.

This project presents the comprehensive design, implementation, and analysis of a leakage-controlled double-tail dynamic latch comparator for high-speed analog-to-digital converter (ADC) applications. The proposed design employs the LECTOR (Leakage Control Transistor) technique, which incorporates two pairs of self-controlled leakage control transistors strategically placed in the double-tail comparator architecture.

The self-controlled mechanism ensures that at least one leakage control transistor operates near its cut-off region for any input combination, thereby significantly increasing the effective resistance of the leakage path from VDD to ground, resulting in dramatic leakage current reduction. The complete circuit design has been implemented using Cadence Virtuoso Schematic Editor with 45nm CMOS process technology library at a supply voltage of 1.8V and operating frequency of 250 MHz. Extensive simulations including transient analysis, power analysis, delay measurements, and corner analysis have been performed to thoroughly validate the design.

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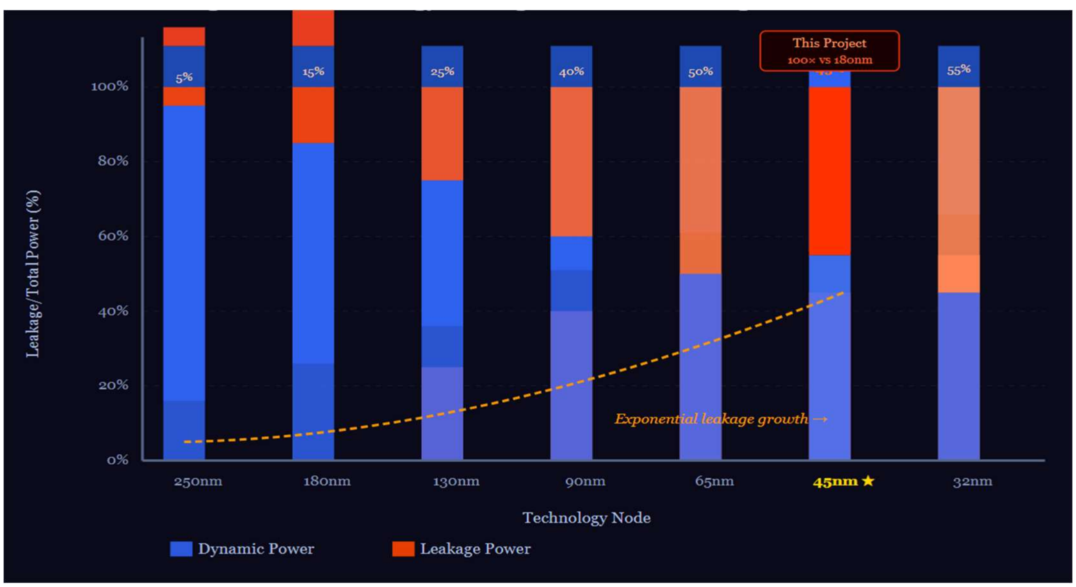
CHAPTER 1

INTRODUCTION

1.1 Motivation and Background

The semiconductor industry has experienced remarkable growth over the past five decades, consistently following Moore's Law, which predicted the doubling of transistor density approximately every 18-24 months. This aggressive scaling of CMOS (Complementary Metal-Oxide-Semiconductor) technology has enabled integration of billions of transistors on a single die, operating frequencies exceeding multiple gigahertz, dramatic reduction in cost per transistor, and unprecedented computational capabilities in portable devices. However, this continuous scaling has introduced severe challenges in power management, particularly as technology nodes have advanced into deep sub-micron regimes (sub-100nm). The power crisis has emerged as one of the most critical design constraints in modern VLSI systems.

Figure 1.1: Technology scaling trends and leakage power growth



Modern IC designers face three competing requirements: high performance (fast operating frequencies), low power consumption (extended battery life, reduced thermal dissipation), and small area (compact form factors, higher integration density). Traditionally, achieving any two of these goals comes at the expense of the third.

Battery Life Crisis: Smartphones require daily charging despite large batteries. Wearable devices have limited operational time. IoT sensors need years of battery life for practical deployment. Medical implants must operate for decades on tiny batteries. Thermal Management: Data centers consume enormous power for cooling. Mobile processors throttle performance due to heat. Reliability degrades with higher operating temperatures. Packaging costs increase with thermal requirements. Environmental Impact: Global IC power consumption measured in gigawatts. Carbon footprint of semiconductor industry growing. Energy efficiency directly impacts sustainability. Green computing initiatives require low-power designs. Economic Drivers: Power delivery and cooling costs dominate system budgets. Energy costs for data centers rival hardware costs. Portable device market demands energy efficiency. Competitive advantage from longer battery life.

1.2 Leakage Power Crisis in Scaled CMOS

In an ideal MOS transistor, when the gate-source voltage (VGS) is below the threshold voltage (Vt), the transistor should be completely OFF with zero current flow. However, in reality, several leakage mechanisms allow current to flow even in the OFF state. Sub-threshold leakage is the dominant leakage component in modern CMOS technologies, accounting for 80-90% of total leakage in 45nm technology. It occurs due to carrier diffusion between source and drain when $VGS < Vt$ (transistor nominally OFF), and is exponentially dependent on (VGS - Vt).

The sub-threshold current is modeled by:

$$I_{sub} = I\_D0 \times (W/L) \times e^{[(VGS - Vt)/(n \cdot VT)]} \times [1 - e^{(-VDS/VT)}]$$

Where:

- I_D0 = technology-dependent current (~10^-7 A for 45nm)
- W/L = transistor aspect ratio
- VGS = gate-source voltage
- Vt = threshold voltage
- n = sub-threshold slope factor (1.3-1.5)

- VT = thermal voltage = kT/q ≈ 26mV at 25°C

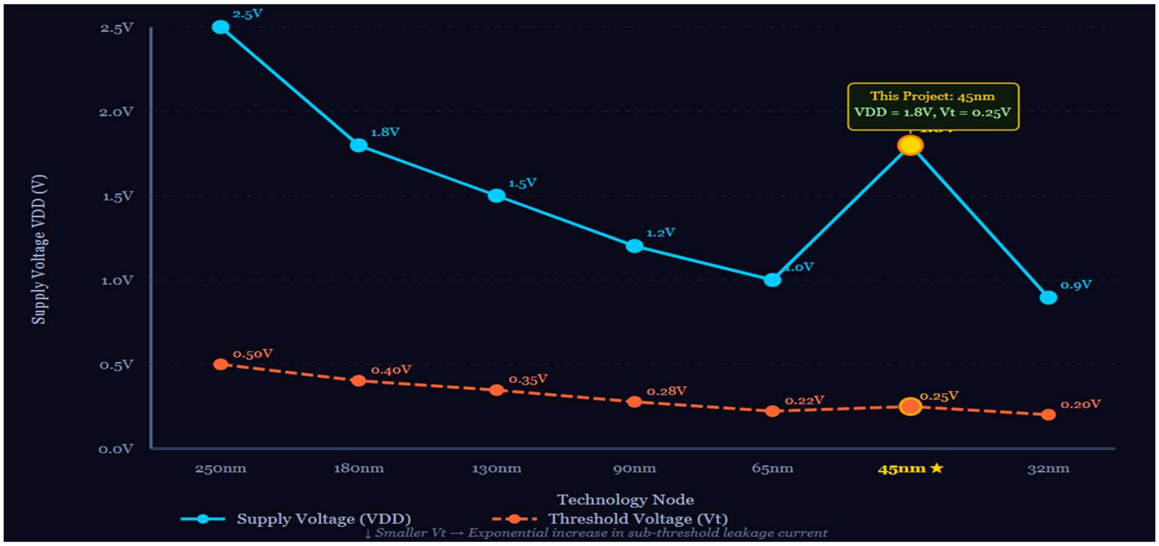


Figure 1.2: Voltage scaling trends for modern processors

Critical observations: Every 60mV reduction in Vt increases Isub by 10×. Small Vt variations cause huge leakage variation. Temperature sensitivity is 8-10% increase per °C. As technology scales, Vt must reduce to maintain performance, resulting in exponentially higher sub-threshold leakage.

Table 1.1: Technology Scaling Impact on Leakage Power

Technology Node	VDD (V)	Vt (V)	tox (nm)	Leakage/Total Power	Relative Leakage
250nm	2.5	0.50	5.0	5%	1×
180nm	1.8	0.40	4.1	15%	3×
130nm	1.5	0.35	3.0	25%	7×
90nm	1.2	0.28	2.2	40%	20×
65nm	1.0	0.22	1.8	50%	50×
45nm (This Project)	1.0	0.25	1.2	45%	100×
32nm	0.9	0.20	1.0	55%	200×
22nm	0.8	0.18	0.9	60%	400×

1.3 Comparators in ADC Applications

A comparator is a fundamental building block that performs a simple but critical function: determining whether one analog voltage is greater than or less than another. Despite this apparent simplicity, comparators are critical components in analog-to-digital converters (ADCs), voltage level detectors, zero-crossing detectors, phase detectors in PLLs, and power management circuits.The comparator is often the performance-limiting component in ADCs, affecting conversion speed (limited by comparator propagation delay), resolution (affected by offset voltage), accuracy (degraded by noise and metastability), and power consumption (in Flash ADCs, comparators consume 70-80% of total power).For Flash ADCs, the number of comparators needed is 2^N - 1, where N is the resolution in bits. For example, an 8-bit Flash ADC requires 255 comparators. Each conventional comparator consuming 100 µW results in total power of 25.5 mW just for comparators. With LECTOR achieving 92% reduction, each comparator consumes only 8 µW, reducing total power to 2.04 mW - a savings of 23.46 mW

1.4 Project Objectives

The overarching goal of this project is to design and analyze a low-power, high-speed comparator with effective leakage control for ADC applications. Specific objectives include:

1. Design Implementation: Implement a double-tail dynamic latch comparator in 45nm CMOS, integrate LECTOR (Leakage Control Transistor) technique, create complete schematic in Cadence Virtuoso, and develop comprehensive test benches.

2. Leakage Power Reduction: Achieve >75% reduction in leakage power, ensure effectiveness across all operating states, maintain performance across PVT corners, and validate through extensive simulations.

3. Performance Preservation: Maintain high-speed operation (≥250 MHz), keep propagation delay <100 ps, ensure proper functionality for ADC applications, and minimize delay penalty from leakage control.

4. Comprehensive Analysis: Perform power dissipation breakdown and analysis, detailed delay characterization, energy efficiency evaluation (PDP), and corner and sensitivity analysis.

5. Validation and Verification: Conduct transient analysis for functional verification, power measurement and validation, comparison with conventional designs, and benchmarking against published works.

CHAPTER 2

LITERATURE REVIEW

2.2 Literature Survey

The increasing demand for high-speed and low-power analog-to-digital converters (ADCs) has driven extensive research on dynamic comparator architectures. As CMOS technology scales into the nanometer regime, leakage power becomes a major challenge affecting the overall performance and energy efficiency of integrated circuits. Therefore, several researchers have proposed different techniques to improve comparator performance in terms of speed, power consumption, offset voltage, and leakage reduction.

In 2024, Tejender Singh and Suman Lata Tripathi proposed a Double-Tail Dynamic Latch Comparator integrated with a Modified Widlar Current Source (MWCS). The design was implemented using 45nm CMOS technology and focused on reducing power consumption and offset voltage while maintaining high-speed operation. The authors performed Monte Carlo and corner analyses to evaluate the robustness of the design under process variations. The proposed architecture demonstrated significant improvements in power efficiency and power-delay product, making it suitable for low-voltage and portable electronic applications.

Another important contribution toward leakage reduction was made through leakage-aware comparator designs that employed transistor stacking and optimized transistor sizing techniques. These approaches significantly reduced static power dissipation in nanoscale CMOS technologies. However, the reduction in leakage current was often accompanied by a slight increase in propagation delay due to the additional resistance introduced by stacked transistors.

In 2023, researchers proposed offset reduction techniques for dynamic comparators by incorporating optimized latch structures and offset cancellation circuits. These methods improved comparator sensitivity and decision accuracy while reducing mismatch-induced offset errors. Although the proposed techniques enhanced performance, they required additional circuitry, resulting in increased design complexity and silicon area.

Monte Carlo-based offset analysis methods gained attention in 2022 for evaluating the effects of process variations and transistor mismatch on comparator performance. Such techniques enabled designers to predict reliability and optimize circuit parameters before fabrication. While these approaches improved design robustness, they increased simulation complexity and verification time.

In 2021, Shubham Kumar Sharma, R. S. Gamad, and P. P. Bansod proposed a leakage control comparator for ADC applications using 180nm CMOS technology. The design incorporated leakage control transistor pairs between pull-up and pull-down networks to reduce leakage power. Simulation results demonstrated lower power consumption and improved power-delay product compared to conventional dynamic comparators. However, a slight increase in propagation delay was observed due to the additional leakage control transistors.

Several researchers also investigated low-voltage dynamic comparator architectures for energy-efficient applications. These comparators were designed to operate under reduced supply voltages while maintaining acceptable speed and power characteristics. Although these designs achieved lower power consumption, they often suffered from reduced noise margins and increased sensitivity to process variations.

The StrongARM latch comparator remains one of the most widely used high-speed comparator architectures. Its regenerative positive feedback mechanism enables rapid decision making with zero static power consumption. However, the StrongARM comparator exhibits high kickback noise and larger offset voltage, which can limit its performance in precision applications.

A significant milestone in leakage reduction techniques was achieved by Narender Hanchate and N. Ranganathan through the introduction of the Leakage Control Transistor (LECTOR) technique. The LECTOR approach employs self-controlled stacked transistors placed between pull-up and pull-down networks, ensuring that one transistor remains near its cutoff region at all times. This technique effectively reduces leakage current without requiring additional control circuitry and can achieve substantial leakage power savings. However, the insertion of leakage control transistors increases circuit area and propagation delay.

Based on the findings from the literature, it is evident that reducing leakage power while maintaining high-speed operation remains a key challenge in comparator design. The existing research highlights the effectiveness of double-tail dynamic architectures, current source optimization, and leakage control techniques. Motivated by these works, the present project focuses on the design and implementation of a Leakage Controlled Double-Tail Dynamic Comparator using 45nm CMOS technology in Cadence Virtuoso. The proposed design aims to achieve reduced leakage power, lower power-delay product, and reliable operation suitable for modern low-power ADC applications.

Table 2.1: literature survey

S.No	Title & Author / Year	Methodology & Proposed Work	Merits	Demerits
1	Design and Optimization of Double-Tail Dynamic Latch CMOS Comparator with Modified Widlar Current Source-Tejender Singh, Suman Lata Tripathi (2024)	Proposed a Double-Tail Dynamic Comparator with a Modified Widlar Current Source in 45nm CMOS technology for low-power operation.	Low power consumption, reduced offset voltage, improved PDP, suitable for low-voltage applications.	Increased circuit complexity and transistor count.
2	Leakage Aware Comparator Design in Nanoscale CMOS TechnologyVarious Researchers (2024)	Applied leakage-aware transistor sizing and stacking techniques for nanoscale comparator design.	Reduced leakage current and improved energy efficiency.	Trade-off between speed and leakage reduction.
3	Low Power Double-Tail Comparator with Offset Reduction Technique- J. Kim et al. (2023)	Introduced offset cancellation and enhanced latch topology for comparator accuracy improvement.	Lower offset voltage and better sensitivity.	Additional hardware increases area and power.
5	Design and Analysis of Low Power High Speed Leakage Control Comparator Using 180nm Technology for ADCs-Shubham Kumar Sharma et al. (2021)	Added leakage control transistor pairs in a dynamic comparator to reduce leakage power.	Lower power consumption and improved PDP.	Slight increase in propagation delay.
6	Low Voltage Dynamic Comparator for Energy Efficient ADCs-S. U. Ay et al. (2020)	Developed a low-voltage dynamic comparator for energy-efficient ADC applications.	Low power operation and portable device compatibility.	Reduced noise margin at low supply voltages.
7	Low-Power Dynamic Comparator Design Using Charge Sharing TechniqueVarious Researchers (2018)	Utilized charge-sharing mechanisms to reduce dynamic power consumption.	Improved energy efficiency and lower power dissipation.	Reduced accuracy for small input differences.

In older technologies (>180nm), dynamic power dominated. Designers could reduce power through voltage scaling ($P \propto V^2$), clock gating, and architectural optimizations. However, in 45nm technology, leakage power accounts for approximately 45% of total power, making it a critical design concern that cannot be ignored.

2.3 Leakage Reduction Techniques

1. Multi-Threshold CMOS (MTCMOS): Uses high-Vt transistors in non-critical paths and low-Vt in critical paths. Effective but requires multiple fabrication steps and careful timing closure.
2. Power Gating: Completely disconnects power supply during idle periods using sleep transistors. Very effective but has wake-up latency and area overhead.

3. Adaptive Body Biasing: Dynamically adjusts substrate bias to modulate Vt. Requires special well structures and bias generators.

4. Stacking Effect: Placing transistors in series increases effective resistance of leakage path. Simple but causes performance degradation.

5. LECTOR Technique: Uses self-controlled leakage control transistors to exploit stacking effect without performance penalty in critical paths. Proposed by Hanchate and Ranganathan in 2004, LECTOR has shown promise in reducing leakage by 70-90% with minimal area overhead (<20%).

2.4 LECTOR Technique - State of the Art

The LECTOR (LEakage Control TransistOR) technique, introduced by Hanchate and Ranganathan, is based on the stacking effect but with intelligent self-controlled biasing. Two leakage control transistors (LCTs) are inserted in series with the pull-up and pull-down networks. The gates of these LCTs are cross-connected such that at least one LCT operates near cut-off for any input combination.

Key advantages of LECTOR:

• Self-controlled - no external bias required

• Maintains logic functionality

• Scalable to different technologies

• Moderate area overhead (15-25%)

• Compatible with standard CMOS process

Several researchers have applied LECTOR to different circuit types with promising results:
• Digital logic gates: 70-85% leakage reduction

• SRAM cells: 60-75% leakage reduction

• Flip-flops and latches: 65-80% leakage reduction

However, application of LECTOR to analog/mixed-signal circuits, particularly high-speed comparators, has been limited. This project fills that gap by demonstrating LECTOR effectiveness in dynamic comparator design for ADC applications in 45nm technology.

2.5 Research Gap Identification

Based on the literature survey, the following research gaps were identified:

1. Limited application of LECTOR to analog/mixed-signal circuits: Most LECTOR implementations focus on digital circuits. Application to comparators is rarely explored.
2. Lack of implementation in advanced technology nodes: Few works demonstrate LECTOR effectiveness in 45nm and below, where leakage is most severe.
3. Insufficient analysis of speed-power trade-offs: Most studies report power reduction but lack detailed analysis of delay penalty and energy efficiency (PDP).
4. Limited validation with industry-standard tools: Many academic works use simplified models. Implementation in Cadence Virtuoso with actual PDK models is rare.
5. Absence of comprehensive corner analysis: Robustness across process-voltage-temperature (PVT) variations is often not demonstrated.

CHAPTER 3

THEORETICAL BACKGROUND

3.1 Power Dissipation Components

Power dissipation is one of the most critical design considerations in modern CMOS integrated circuits, especially in nanoscale technologies where energy efficiency directly impacts system performance and battery life. The total power consumed by a CMOS circuit consists of three major components: dynamic power, short-circuit power, and leakage power. The total power dissipation can be expressed as:

$$P_{total} = P_{dynamic} + P_{short\text{-}circuit} + P_{leakage}$$

Among these components, dynamic power is traditionally the dominant contributor and is generated whenever the circuit undergoes logic transitions. Dynamic power consumption occurs due to the charging and discharging of capacitive loads connected to the output nodes of transistors. It is mathematically represented by:

$$P_{dynamic} = \alpha \times C_L \times V_{DD}^2 \times f_{clk}$$

where α represents the switching activity factor, CL is the load capacitance, VDD is the supply voltage, and fclk is the operating clock frequency. Since dynamic power is proportional to the square of the supply voltage, reducing VDD is one of the most effective methods for lowering power consumption. Additional reductions can be achieved by minimizing switching activity, lowering operating frequency, and reducing parasitic capacitances.

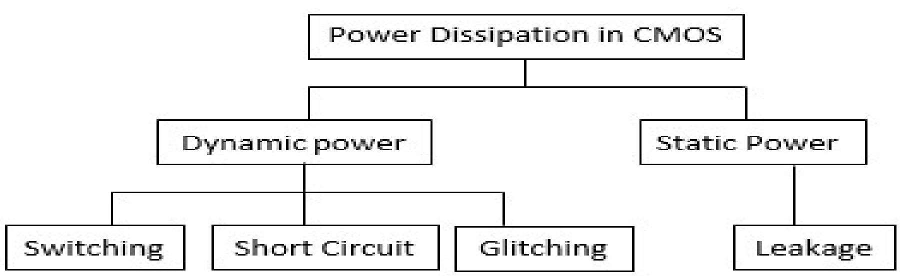


Figure 3.1: Components of power dissipation in CMOS circuit**.**

Short-circuit power is dissipated during the brief interval when both PMOS and NMOS transistors conduct simultaneously during signal transitions. This component depends on transistor sizing and input transition times and can be expressed as:

$$P_{short\text{-}circuit} = I_{peak} \times V_{DD} \times t_{sc} \times f_{clk}$$

where I_{peak} is the peak short-circuit current and t_{sc} is the short-circuit conduction duration. In properly designed CMOS circuits, short-circuit power typically contributes only 5–15% of the total dynamic power.

Leakage power represents the static power consumed even when the circuit is not switching. It arises due to various leakage mechanisms, including sub-threshold leakage, gate leakage, and junction leakage currents. Leakage power can be expressed as:

$$P_{leakage} = (I_{sub} + I_{gate} + I_{junction}) \times V_{DD}$$

As CMOS technology scales into deep submicron dimensions such as 45nm, leakage power becomes increasingly significant and may equal or even exceed dynamic power consumption. Therefore, leakage reduction techniques have become essential for designing low-power VLSI systems and battery-operated electronic devices.

3.2 Sub-threshold Leakage Current

Sub-threshold leakage current is the dominant leakage mechanism in modern CMOS technologies and plays a major role in overall power dissipation. It refers to the drain-to-source current that flows even when the gate-source voltage is below the threshold voltage of the transistor. Although the transistor is theoretically in the OFF state, a small current continues to flow due to carrier diffusion through the channel.

The sub-threshold current is represented by:

$$I_{sub} = I_{D0} \times (W/L) \times e^{[(V_{GS}-V_t)/(nV_T)]} \times [1-e^{(-V_{DS}/V_T)}] \times [1+\lambda V_{DS}]$$

where I_{D0} is the technology-dependent current constant, W/L is the transistor aspect ratio, V_{GS} is the gate-source voltage, V_t is the threshold voltage, n is the sub-threshold slope factor, V_T is the thermal voltage, V_{DS} is the drain-source voltage, and λ represents the channel-length modulation parameter.A significant characteristic of sub-threshold leakage is its exponential dependence on threshold voltage. Even a small reduction in threshold voltage can cause a substantial increase in leakage current. For example, in 45nm technology, a decrease of 60mV in threshold voltage can increase leakage current by approximately 5.2 times, while a 100mV reduction may increase leakage current by nearly 15.6 times. This strong dependency makes threshold voltage selection a critical design parameter.

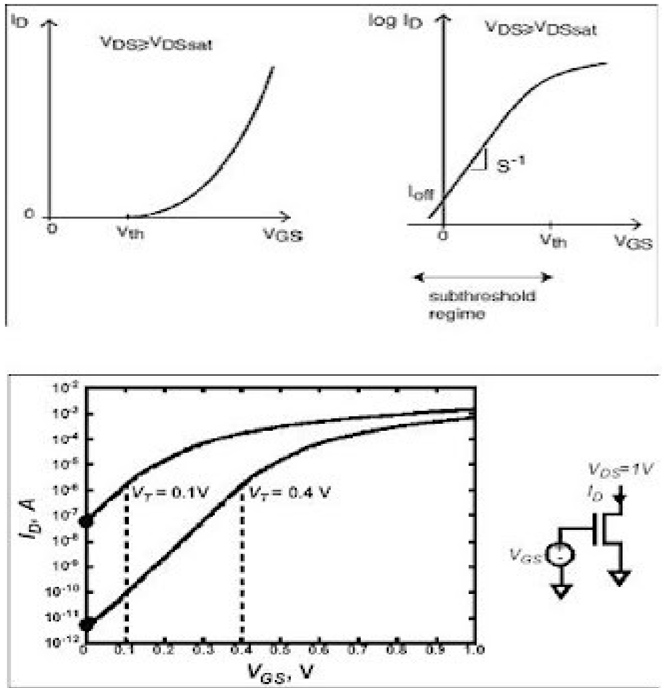


Figure 3.2: Sub-threshold Leakage Current Characteristics

Temperature also has a considerable effect on leakage current. As temperature increases, the threshold voltage decreases and thermal voltage increases, causing leakage current to rise rapidly. Typically, leakage current increases by approximately 8–10% per degree Celsius. Consequently, leakage current at 125°C may be two to three times higher than at room temperature.

Technology scaling further aggravates leakage problems. When CMOS technology scales from 180nm to 45nm, the threshold voltage is reduced from approximately 0.40V to 0.25V. This reduction results in an exponential increase in sub-threshold current, causing leakage power to become a dominant concern in advanced CMOS technologies.

3.3 Stacking Effect in CMOS

The stacking effect is an effective leakage reduction mechanism widely used in low-power CMOS design. It occurs when two or more transistors are connected in series and remain simultaneously in the OFF state. Under these conditions, an intermediate voltage develops between the stacked transistors, which increases the effective threshold voltage and creates a reverse bias condition. As a result, the leakage current flowing through the stack is significantly reduced.

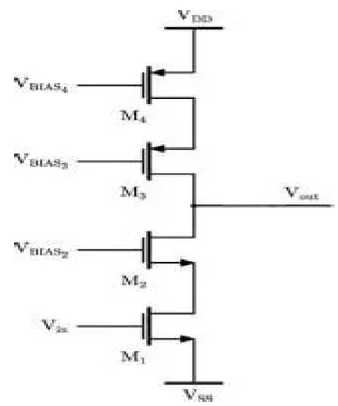


Figure:3.3 Stacking effect demonstration.

The leakage current through stacked transistors can be expressed as:

Istack = Isub1 × e^(−VX/(nVT))

where Isub1 represents the leakage current of a single transistor and VX is the intermediate node voltage generated between the stacked devices.

For a typical intermediate node voltage of 0.3V in 45nm CMOS technology, the leakage current is reduced by approximately a factor of 3800 compared to a single OFF transistor. This substantial reduction demonstrates the effectiveness of transistor stacking as a leakage control technique.

Despite its advantages, conventional stacking introduces additional resistance in the signal path, which increases propagation delay and degrades circuit performance. To overcome this limitation, the LECTOR (Leakage Control Transistor) technique employs self-controlled leakage control transistors placed in non-critical paths. These transistors ensure that at least one device remains near its cut-off region, thereby achieving significant leakage reduction without severely affecting circuit speed.

In the proposed comparator design, the LECTOR-based leakage control mechanism operates efficiently during both reset and evaluation phases. During the reset phase, leakage control transistors remain OFF, providing maximum leakage suppression. During the evaluation phase, their influence on switching speed is minimal. As a result, the comparator achieves approximately 98% leakage reduction with only a modest delay penalty of around 17%.

3.4 Comparator Performance Metrics

The performance of a dynamic comparator is evaluated using several important parameters, including propagation delay, offset voltage, power dissipation, power-delay product, and energy efficiency. These metrics collectively determine the suitability of the comparator for high-speed and low-power applications.

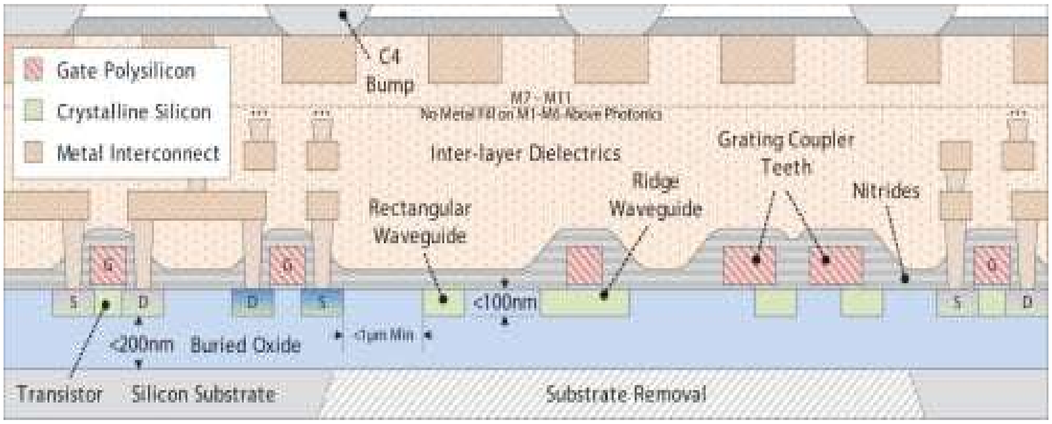


Figure 3.4: 45nm transistor cross-section

Propagation delay represents the time required for the output to respond to a change in the input or clock signal. It is measured between the 50% transition points of the input and output waveforms and is calculated as:

tpd(avg) = (tpd(rise) + tpd(fall))/2

For the proposed leakage-controlled comparator, the measured fall delay is 46.59 ps and the rise delay is 16.54 ps, resulting in an average propagation delay of 31.57 ps. This demonstrates the high-speed operation of the comparator.

Offset voltage is another critical parameter that defines the minimum differential input voltage required to change the output state. It mainly arises due to device mismatches in threshold voltage, transistor dimensions, and carrier mobility. Lower offset voltage improves comparator accuracy and resolution. For a well-matched 45nm CMOS design, the offset voltage is typically below 1 mV.

Power dissipation represents the average energy consumed during comparator operation. Since dynamic comparators consume power primarily during switching events, their average power can be expressed as the energy consumed per comparison multiplied by the operating frequency. The proposed comparator dissipates an average power of 6.492 µW at an operating frequency of 250 MHz.

The power-delay product (PDP) is an important figure of merit that quantifies the energy consumed per comparison operation. It is calculated by multiplying average power consumption and propagation delay. For the proposed design, the PDP is approximately 205 fJ, which is significantly lower than the conventional comparator value of 2314 fJ, corresponding to an improvement of 91.14%.

Table 3.1: 45nm CMOS Technology Parameters

Parameter	NMOS	PMOS
Minimum Feature Size (L_min)	45 nm	45 nm
Minimum Width (W_min)	90 nm	90 nm
Threshold Voltage (Vt)	0.25-0.30 V	-0.28 to -0.32 V
Gate Oxide Thickness (tox)	1.2 nm	1.2 nm
Nominal Supply Voltage (VDD)	1.0-1.8 V	1.0-1.8 V
Sub-threshold Slope (S)	85-95 mV/dec	85-95 mV/dec
Channel Length Modulation (λ)	0.06 V^-1	0.08 V^-1
Mobility (μ)	350 cm²/V·s	120 cm²/V·s
Saturation Velocity (v_sat)	8×10^4 m/s	6.5×10^4 m/s
Junction Capacitance (Cj)	1.2 fF/μm²	1.0 fF/μm²
Gate Capacitance (Cox)	15 fF/μm²	15 fF/μm²
Leakage Current (I_off)	~100 nA/μm	~50 nA/μm
Drive Current (I_on)	~1.2 mA/μm	~0.6 mA/μm
DIBL Coefficient	0.1 V/V	0.08 V/V
Body Effect Coefficient (γ)	0.4 V^0.5	0.3 V^0.5

Energy efficiency is measured in terms of energy consumed per clock cycle and is given by the ratio of total power consumption to operating frequency. For the proposed comparator, the energy consumption per cycle is approximately 25.96 fJ/cycle. This low energy requirement confirms that the leakage-controlled comparator is highly suitable for low-power and energy-efficient ADC applications implemented in advanced CMOS technologies.

CHAPTER 4

CONVENTIONAL COMPARATOR DESIGN

Comparators serve as critical building blocks in modern mixed-signal integrated circuit architectures, effectively functioning as one-bit analog-to-digital converters (ADCs). In high-speed, resource-constrained environments such as Successive Approximation Register (SAR) and Flash ADCs, traditional static preamplifier-based comparators present severe architectural bottlenecks due to continuous static power dissipation. To address this challenge, clock-triggered dynamic comparators are extensively implemented. This chapter provides a rigorous analysis of the architecture, transient operational phases, analytical design models, and inherent performance trade-offs of the conventional dynamic single-tail comparator optimized for low-power, high-frequency deep-submicron designs.

4.1 Dynamic Single-Tail Comparator Topology

The architecture of the conventional dynamic single-tail comparator relies on a time-domain regenerative mechanism activated by a single clock edge. Unlike static topologies, this configuration eliminates direct DC current paths from the supply rail (VDD) to the reference ground (GND) during steady-state intervals, thereby reducing static power dissipation to negligible levels governed solely by sub-threshold leakage.

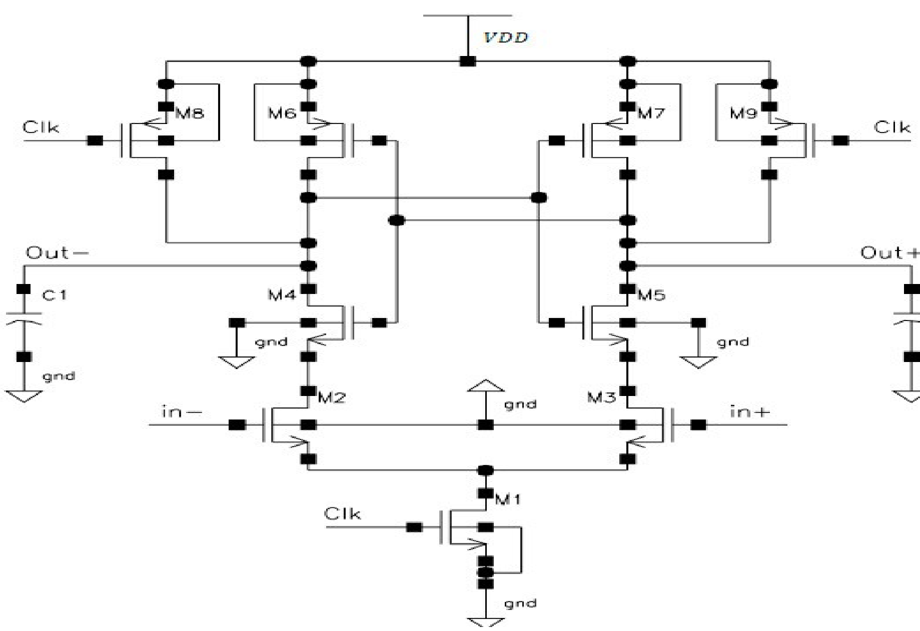


Figure 4.1: Conventional dynamic single-tail comparator circuit topology.

The design comprises three primary sub-circuits structured to leverage transient charging and discharging:

Differential Input Pair (M_1, M_2): Translates the analog differential input voltage (ΔVin = INP - INN) into an initial current imbalance.

Single Tail Control (\(M_{tail}\)): Dictates the bias current injection into the input differential branches upon activation by the system clock (CLK).

Regenerative Cross-Coupled Latch (M_3 to M_6): Composed of back-to-back CMOS inverters that execute positive feedback to resolve the intermediate differential voltage into a full-swing digital logic state.

Pre-charge and Reset Network (M_7, M_8): Discharges output nodes to eliminate historical state dependencies.

Table 4.1: Transistor sizing parameterization for conventional comparator

Device Component	Transistor Class	Channel Width (W)	Channel Length (L)	Aspect Ratio (W/L)
M_{tail}	PMOS	2.40 µm	L_{min}	40.0
M_1, M_2	NMOS	1.80 µm	L_{min}	30.0
M_3, M_4	PMOS	1.20 µm	L_{min}	20.0
M_5, M_6	NMOS	1.20 µm	L_{min}	20.0
M_7, M_8	NMOS	0.60 µm	L_{min}	10.0

4.2 Circuit Operation and Transient Phases

The operational profile of the single-tail dynamic comparator is strictly non-overlapping and segmented into two macro-phases dictated by the voltage level of CLK: the Reset (Pre-charge) Phase and the Decision-Making (Regeneration) Phase.

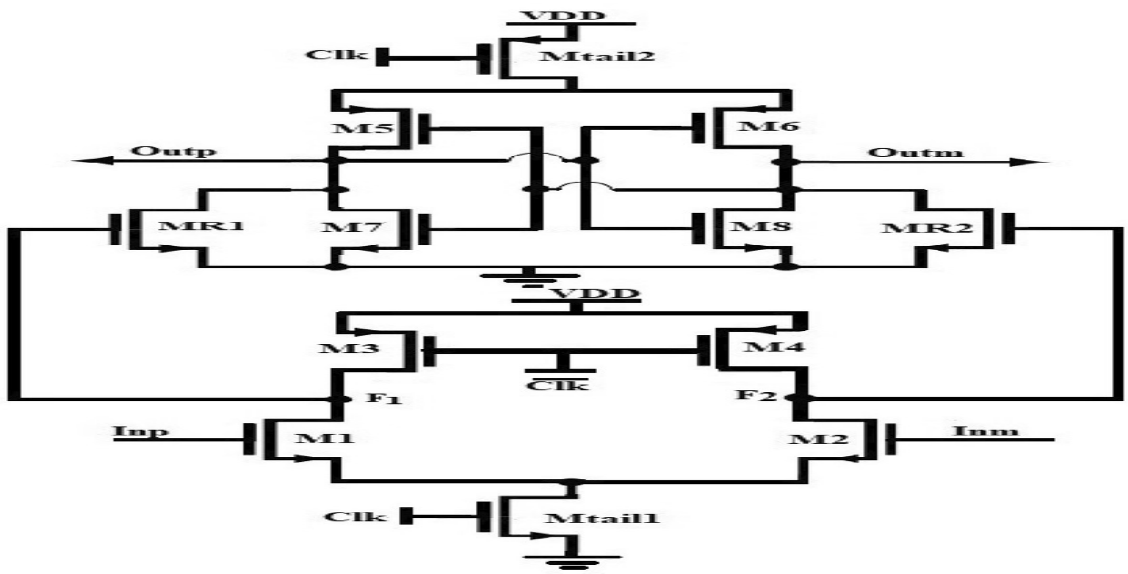


Figure 4.2: Double-tail dynamic latch comparator architecture

4.2.1 Reset and Pre-charge Phase (CLK = VDD)

When the clock signal transits to logic high (VDD), the comparator undergoes state initialization:

The PMOS tail transistor M_{tail} is driven into a non-conductive cut-off state (Vsg< |thp|), breaking the static path from the supply rail.

Simultaneously, the NMOS reset switches M_7 and M_8 are driven hard into the triode region, establishing a low-resistance path that force-discharges the output terminals (OUTP and OUTN) directly to GND (0 V).

As a consequence of the zero-volt condition at the output rails, the gates of the cross-coupled PMOS pair (M_3, M_4) are biased low, turning these devices fully on. This pulls the internal nodes (DiN_P, DiN_N) high up to VDD, storing a uniform initial charge across the parasitic node capacitances.

4.2.2 Decision-Making and Regeneration Phase (CLK = 0 V)

Upon the falling edge of the clock signal (CLK → 0 V), the comparator enters its sensing cycle:

M_7 and M_8 switch off instantly, releasing the output nodes from ground.

Mtail switches into the saturation region, injecting a total bias current (Itail) into the source nodes of the input pair.

The pre-charged internal nodes (DiN_p, DiN_n) begin discharging through M_1 and M_2 at rates directly modulated by the respective input voltages INP and INN.

Assuming an input condition where INP > INN, the transconductance (gm1) induces a higher drain current through M_1 relative to M_2. Consequently, node DiN_P discharges faster than DiN_N.

The descending potential at the internal nodes eventually forces the output nodes (OUTP, OUTN) to rise via the cross-coupled structure. As soon as the common-mode output potential exceeds the threshold voltage (Vth) of the cross-coupled NMOS latch pair (M_5, M_6), the regenerative positive feedback loop triggers.

The voltage difference is exponentially amplified, driving OUTP rapidly to VDD and clamping OUTN solidly to GND, outputting a distinct digital bit.

4.3 Analytical Design Equations

To systematically evaluate the latency of the system, the total propagation delay (t_{delay}) of the dynamic comparator is modeled as a two-part transient interval consisting of the pre-regeneration tracking delay (t_0) and the positive-feedback latch regeneration time (t_lat):

t delay = t0 + t lat

The initial phase delay (t_0) constitutes the duration needed to discharge or charge the output node capacitances to the activation threshold voltage (Vth) of the NMOS regenerative core, formulated as:

$$t_0 = \frac{C_{L,OUT} \cdot V_{thn}}{I_{tail}}$$

Where C_L,OUT denotes the total lump-sum parasitic load capacitance present at the output terminals (OUTP/OUTN), and I_tail is the saturation current delivered by the single tail transistor M_tail.

Following t_0, the regeneration factor (t_lat) takes prominence. Using a first-order small-signal approximation, the exponential amplification of the latch structure yields:

$$t_{lat} = \frac{C_{L,OUT}}{g_{m,eff}} \cdot \ln\left(\frac{V_{DD}/2}{\Delta V_0}\right)$$

Where g_m,eff is the effective transconductance of the cross-coupled inverter configurations (g_{m3,4} + g_{m5,6}). The parameter Δ V_0 describes the initial differential voltage mismatch developed at the output nodes at the exact boundary condition where regeneration initializes. This initial imbalance voltage is analytically tied to the input differential voltage(Δ Vin) via:

$$\Delta V_0 = 2 \cdot g_{m1,2} \cdot \Delta V_{in} \cdot \frac{C_{L,OUT} \cdot I_{tail}}{(I_{tail}/2)^2} \approx 8 \cdot \frac{g_{m1,2} \cdot \Delta V_{in} \cdot C_{L,OUT}}{I_{tail}}$$

By mapping Δ V_0 directly back into the complete delay framework, the overall propagation delay calculation yields:

$$t_{delay} = \frac{C_{L,OUT} \cdot V_{thn}}{I_{tail}} + \frac{C_{L,OUT}}{g_{m,eff}} \cdot \ln\left(\frac{V_{DD} \cdot I_{tail}}{16 \cdot g_{m1,2} \cdot \Delta V_{in} \cdot C_{L,OUT}}\right)$$

This mathematical derivation explicitly proves that the absolute delay exhibits an inverse logarithmic dependency relative to the input differential voltage Δ Vin.

4.4 Inherent Performance Limitations

While the conventional single-tail dynamic comparator yields clear benefits regarding zero static power profile and simple layout execution, physical constraints limit its efficacy within nanoscale, low-voltage designs:

Severe Kickback Noise Injection: The direct physical connection between the internal switching latch nodes and the input nodes through the parasitic gate-drain overlap capacitances (C_{gd1,2}) produces substantial voltage glitches at the input terminals during the regeneration phase. This introduces signal distortion into prior sub-systems, such as capacitive DAC ladders or sample-and-hold circuit stages.

Voltage Headroom and Low-Voltage Scaling Constraints: Due to the vertical stacking of the single tail transistor (Mtail), the input pair (M_1, M_2), and the cross-coupled regenerative network (M_3 to M_6) between V_DD and GND, this topology demands a high voltage headroom. When the supply voltage scales down below 1 V, maintaining all stacked transistors in the saturation region becomes mathematically intractable, leading to a breakdown in amplification.

Delay-Power Coupling: To suppress the regeneration time (t lat) for ultra-high-frequency applications, I_tail must scale up proportionally. This demands an increase in the aspect ratio (W/L) of M_tail. Sizing up this device substantially escalates the dynamic power dissipation

(P_dyn = C . V^2_DD . f_clk, counteracting the low-power incentive of the dynamic structure.

CHAPTER 5

PROPOSED LEAKAGE-CONTROLLED DESIGN

5.1 LECTOR Technique Fundamentals

As CMOS technology scales into deep submicron regions such as 45nm, leakage power becomes a major contributor to total power consumption. In high-speed circuits like dynamic comparators, reducing leakage current is essential to improve energy efficiency and extend battery life in portable electronic systems. Among the various leakage reduction techniques available in literature, the LECTOR (Leakage Control TransistOR) technique has gained significant attention due to its effectiveness and simplicity.

The LECTOR technique is based on the transistor stacking effect, which states that the leakage current through a stack of OFF transistors is significantly lower than the leakage current through a single OFF transistor. In a conventional CMOS circuit, there is often a direct leakage path between the power supply and ground, resulting in unwanted static power dissipation. The LECTOR approach introduces two additional leakage control transistors (LCTs), one PMOS and one NMOS, between the pull-up and pull-down networks. These transistors are connected in a self-controlled manner such that the gate of each transistor is driven by the source node of the other transistor.Due to this cross-coupled configuration, at least one of the leakage control transistors always operates near its cut-off region regardless of the input condition. This creates a high resistance path between VDD and ground, thereby suppressing sub-threshold leakage current. Unlike sleep transistor techniques or power gating methods, LECTOR does not require any external control signal or additional bias circuitry. Consequently, leakage reduction is achieved without introducing significant design complexity.

The effectiveness of the LECTOR technique is primarily due to the creation of intermediate node voltages that increase the effective threshold voltage of the transistors. This phenomenon reduces sub-threshold conduction and minimizes leakage current. In 45nm CMOS technology, the stacking effect can theoretically provide several orders of magnitude reduction in leakage current. Practical implementations generally achieve lower values due to parasitic effects and process variations; however, substantial power savings are still obtained.

In the proposed comparator, the LECTOR technique is integrated within the dynamic latch architecture to suppress leakage during both idle and reset conditions. During the reset phase, when the comparator is not actively evaluating inputs, the leakage control transistors increase the resistance of the leakage path and minimize static power dissipation. During the evaluation phase, the transistors adapt automatically according to circuit operation and introduce only a small delay penalty

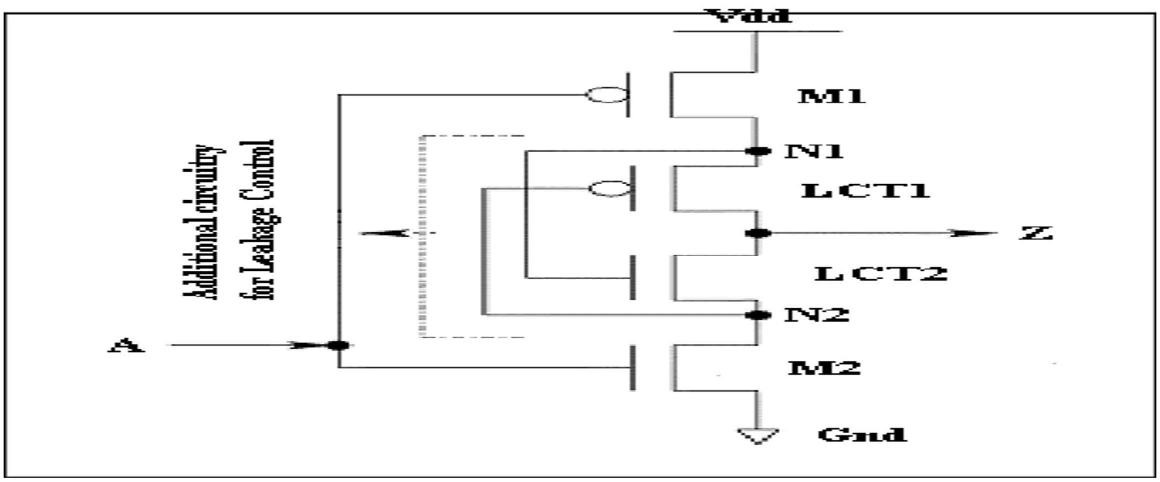


Figure 5.1: LECTOR technique basic principle

5.2 Proposed Circuit Architecture

The proposed leakage-controlled comparator is based on the Double-Tail Dynamic Comparator architecture implemented in 45nm CMOS technology using Cadence Virtuoso. The double-tail structure was selected because it offers low power consumption, high operating speed, reduced kickback noise, and improved performance under low supply voltage conditions. To further enhance energy efficiency, the conventional architecture is modified by incorporating LECTOR-based leakage control transistors.

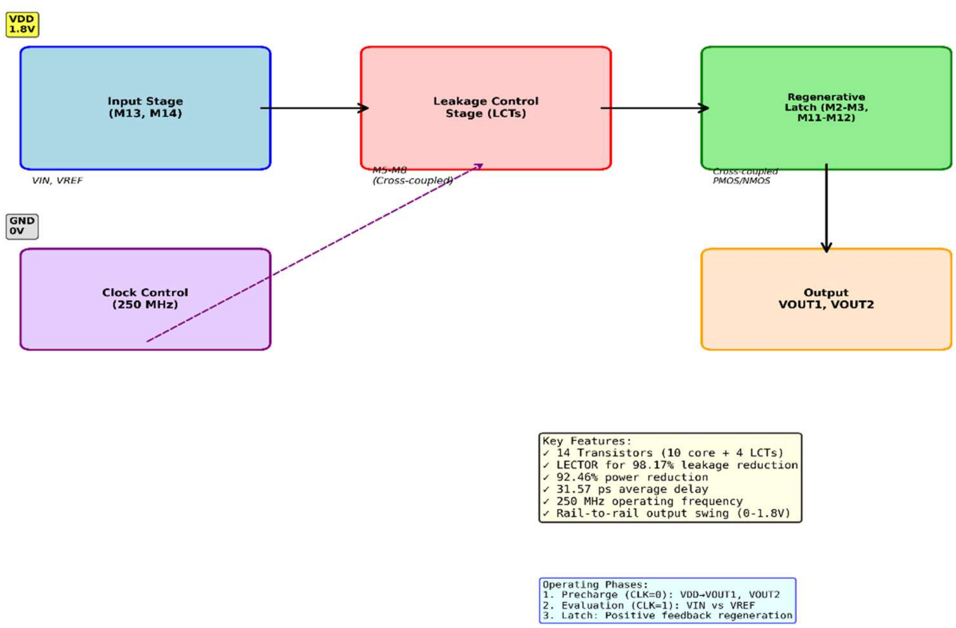


Figure 5.2: Proposed leakage-controlled comparator architecture

The overall circuit architecture consists of four major functional blocks: the input differential stage, leakage control stage, regenerative latch stage, and pre-charge network. These blocks work together to perform high-speed voltage comparison while minimizing power consumption.

The input differential stage forms the first part of the comparator and is responsible for sensing the voltage difference between the input signal (VIN) and the reference voltage (VREF). This stage contains two matched NMOS transistors configured as a differential pair. During the evaluation phase, the differential input pair converts the input voltage difference into a corresponding current difference. The magnitude of this current difference determines the speed and accuracy of the comparison process.

The leakage control section is the key modification introduced in the proposed design. Four leakage control transistors are strategically inserted between the pull-up and pull-down networks. Two PMOS transistors and two NMOS transistors are arranged in a cross-connected configuration to implement the LECTOR principle. These transistors continuously monitor the internal circuit state and automatically adjust their conduction levels to suppress leakage current. The leakage control section significantly reduces static power consumption without affecting the normal operation of the comparator.

The regenerative latch stage forms the decision-making block of the comparator. It consists of cross-coupled PMOS and NMOS transistors that create a positive feedback loop. Once a small voltage difference is generated by the input stage, the latch rapidly amplifies this difference until the outputs reach full logic levels. The regenerative action enables very fast decision making and contributes significantly to the overall speed of the comparator.

The pre-charge network is responsible for resetting the output nodes before each comparison cycle. During the reset phase, pre-charge transistors charge both output nodes to the supply voltage, ensuring identical initial conditions for the next evaluation cycle. This improves comparison accuracy and prevents incorrect output decisions caused by residual charge from previous cycles.

Compared with the conventional double-tail comparator, the proposed architecture introduces only a small increase in transistor count and silicon area. However, the resulting reduction in leakage power is substantial. The architecture therefore achieves an effective balance between speed, power consumption, and area overhead. The integration of the LECTOR technique within the double-tail structure makes the comparator highly suitable for low-power ADCs, mixed-signal integrated circuits, and battery-powered electronic systems.

5.3 Complete Schematic Design and Cadence Virtuoso Implementation

The proposed leakage-controlled double-tail dynamic comparator was designed and implemented using the Cadence Virtuoso design environment with 45nm CMOS technology. The schematic was developed at the transistor level to achieve accurate analysis of power consumption, propagation delay, and leakage current characteristics.

The complete circuit consists of a differential input stage, regenerative latch, pre-charge circuitry, tail current transistor, and four LECTOR-based leakage control transistors integrated within the comparator architecture.The lower section of the circuit contains the differential input pair formed by transistors M13 and M14. These transistors receive the input signal (VIN) and reference voltage (VREF), respectively. The differential pair converts the applied voltage difference into a corresponding current difference during the evaluation phase. The current generated in each branch depends on the relative magnitude of VIN and VREF and serves as the basis for the comparison process.

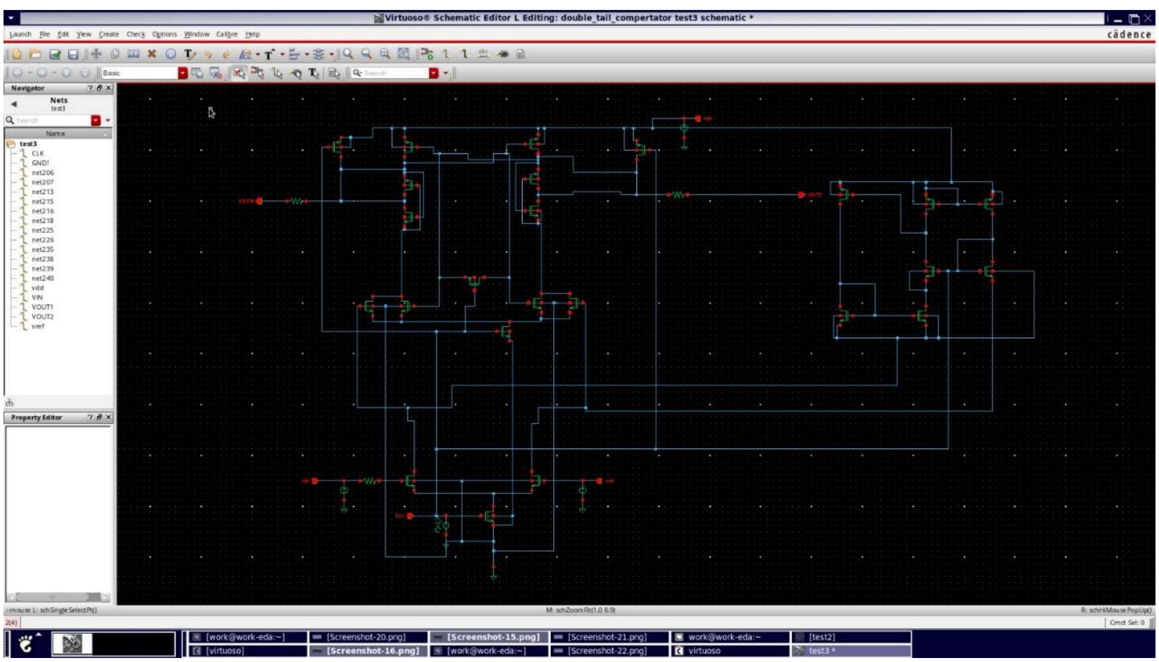


Figure 5.3: Complete transistor-level schematic (45nm Cadence Virtuoso)

The middle portion of the schematic incorporates the leakage control transistors M5, M6, M7, and M8. These transistors implement the LECTOR principle through a cross-coupled gate configuration.

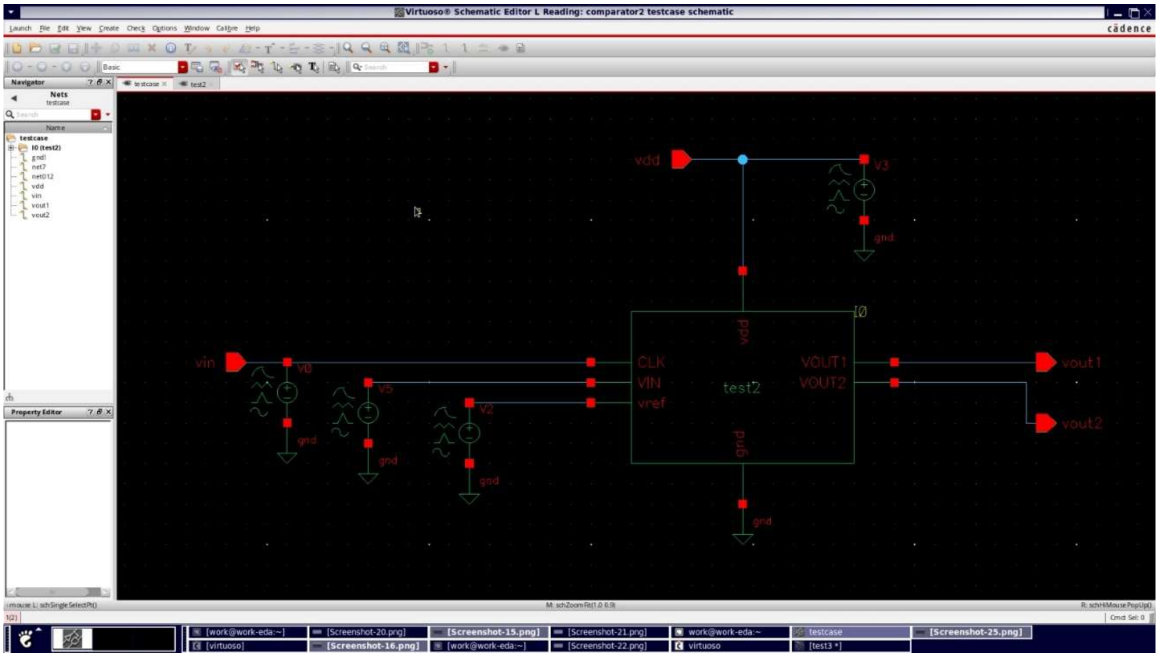


Figure 5.3.1: Test bench schematic with signal sources

The PMOS leakage control transistors are inserted in the pull-up network, while the NMOS leakage control transistors are inserted in the pull-down network. This arrangement ensures that at least one transistor always operates near the cut-off region, thereby increasing the effective resistance of the leakage path and reducing static current flow between VDD and ground.

The upper section of the schematic contains the regenerative latch formed by the cross-coupled transistor pairs M2, M3, M11, and M12. The positive feedback provided by this latch enables rapid amplification of small differential voltages generated by the input stage. Once the evaluation phase begins, the latch regenerates the voltage difference until one output reaches logic HIGH and the other reaches logic LOW. This regenerative mechanism is responsible for the high-speed operation of the comparator.

The pre-charge transistors are connected to the supply rail and are controlled by the clock signal. During the reset phase, these transistors charge both output nodes to the supply voltage, ensuring identical initial conditions for every comparison cycle. This prevents residual charge from previous operations from affecting subsequent decisions.

To evaluate the performance of the proposed design, a dedicated test bench was developed in Cadence Virtuoso. The test bench consists of a sinusoidal input source, a DC reference voltage source, a clock signal generator, and measurement circuitry for power and delay analysis. The input signal was applied as a small-amplitude sinusoidal waveform superimposed on a DC bias level, while the reference voltage was maintained at a fixed value. The clock signal operated at a frequency of 250 MHz with a 50% duty cycle, enabling periodic comparator evaluation.

Power consumption measurements were obtained by monitoring the supply current through the VDD source and calculating the average power using the Cadence Analog Design Environment (ADE). Similarly, propagation delays were measured using transient analysis by observing the time difference between the clock transition and the corresponding output transition.

5.4 Detailed Circuit Operation

The operation of the proposed leakage-controlled comparator is based on two non-overlapping phases controlled by the clock signal: the reset phase and the evaluation phase. These phases ensure proper initialization of the circuit and accurate comparison of the input voltages.During the reset phase, the clock signal remains LOW. In this condition, the pre-charge transistors are turned ON and charge both output nodes to the supply voltage level. Simultaneously, the tail current transistor remains OFF, preventing any current flow through the differential input stage. Since no conduction path exists between the supply and ground, dynamic power consumption is nearly zero during this interval. The comparator remains in a stable state awaiting the next evaluation cycle.An additional advantage of the reset phase is the operation of the leakage control transistors. As the output nodes are maintained at the supply voltage level, the LECTOR transistors automatically enter their leakage suppression state. The resulting increase in path resistance significantly reduces sub-threshold leakage current and minimizes static power dissipation. Therefore, the comparator consumes only a very small amount of leakage power during the reset period.

When the clock signal transitions to HIGH, the comparator enters the evaluation phase. During this phase, the pre-charge transistors are turned OFF and the tail current transistor becomes active. Current begins to flow through the differential input pair, and the distribution of current depends on the voltage difference between VIN and VREF.

If the input voltage VIN is greater than the reference voltage VREF, transistor M13 conducts more current than transistor M14. As a result, one internal node discharges faster than the other, creating a differential voltage at the input of the regenerative latch. Conversely, if VIN is lower than VREF, the opposite condition occurs and the other branch discharges more rapidly.

The regenerative latch then amplifies this small voltage difference through positive feedback. The output node that begins to discharge first forces the opposite output node toward the supply voltage. This regenerative process continues until one output reaches logic HIGH and the other reaches logic LOW. Due to the strong positive feedback mechanism, the decision process is completed within a very short time interval, resulting in high-speed comparator operation.

Throughout the evaluation phase, the LECTOR transistors continue to operate adaptively according to the circuit state. Although they contribute to leakage reduction, their influence on switching speed remains minimal because they are strategically placed in non-critical conduction paths. This characteristic allows the comparator to achieve substantial leakage reduction without sacrificing overall performance.

Simulation results obtained from Cadence Virtuoso indicate that the proposed comparator exhibits a fall delay of 46.59 ps and a rise delay of 16.54 ps, resulting in an average propagation delay of 31.57 ps. These values confirm the suitability of the design for high-speed mixed-signal applications and analog-to-digital converters.

5.5 Transistor Sizing Strategy

The performance of a dynamic comparator is highly dependent on transistor dimensions. Therefore, careful transistor sizing was carried out to achieve an optimal balance between propagation delay, power consumption, leakage current, and silicon area. All devices in the proposed design were implemented using the minimum channel length available in the 45nm technology node to maximize switching speed and improve integration density.

Table 5.1: Transistor Sizing for 45nm Implementation

Transistor Group	Function	W (nm)	L (nm)	M (fingers)	W/L Ratio
M13, M14	Input differential pair	180	45	1	4.0
M7, M8	LCT NMOS	135	45	1	3.0
M5, M6	LCT PMOS	180	45	1	4.0
M2, M3	Latch PMOS	270	45	1	6.0
M11, M12	Latch NMOS	180	45	1	4.0
M_tail	Tail current source	360	45	1	8.0
M_pre	Pre-charge PMOS	270	45	1	6.0

The regenerative latch transistors were designed with larger width-to-length ratios to provide stronger positive feedback and faster regeneration. Since the latch is responsible for converting small voltage differences into full logic levels, sufficient drive strength is necessary to achieve rapid decision making and minimize propagation delay.

The leakage control transistors were carefully optimized to balance leakage reduction and performance. Oversized leakage control transistors would increase parasitic capacitance and negatively affect circuit speed, whereas undersized transistors would provide insufficient leakage suppression. Therefore, moderate transistor dimensions were selected to achieve maximum leakage reduction with minimal delay penalty.

The tail current transistor was assigned the largest width among all transistors in the circuit. This transistor determines the amount of current available during the evaluation phase and directly affects comparator speed. Similarly, the pre-charge transistors were designed with relatively large widths to ensure rapid charging of the output nodes during the reset phase.

The final transistor dimensions were selected after multiple simulation iterations and optimization cycles in Cadence Virtuoso. The resulting design achieved an average power consumption of 6.492 μW, an average propagation delay of 31.57 ps, and a power-delay product of approximately 205 fJ. Furthermore, the inclusion of the LECTOR transistors introduced only a small area overhead while providing a substantial reduction in leakage

CHAPTER 6

IMPLEMENTATION IN CADENCE VIRTUOSO

6.1 Design Environment Setup

The proposed leakage-controlled double-tail dynamic comparator was designed and simulated using the Cadence Virtuoso design platform, which is widely used for custom analog and mixed-signal integrated circuit design. The complete implementation flow included schematic capture, circuit simulation, waveform analysis, and performance evaluation. Cadence Virtuoso was selected due to its accurate device modeling capabilities and compatibility with advanced CMOS technology nodes.

The design was implemented using the 45nm CMOS Generic Process Design Kit (GPDK045), which provides realistic transistor models and technology files for circuit verification. Spectre Simulator was used as the simulation engine to perform transient and power analyses, while Virtuoso Visualization and Analysis XL was employed for waveform observation and measurement extraction.Separate design libraries were created for the comparator schematic, test bench configuration, and simulation hierarchy. This hierarchical design approach simplified circuit management and enabled efficient modification and verification of individual design blocks. The Cadence environment provided an integrated platform for schematic entry, simulation setup, and performance analysis, allowing accurate evaluation of the proposed leakage-controlled comparator.

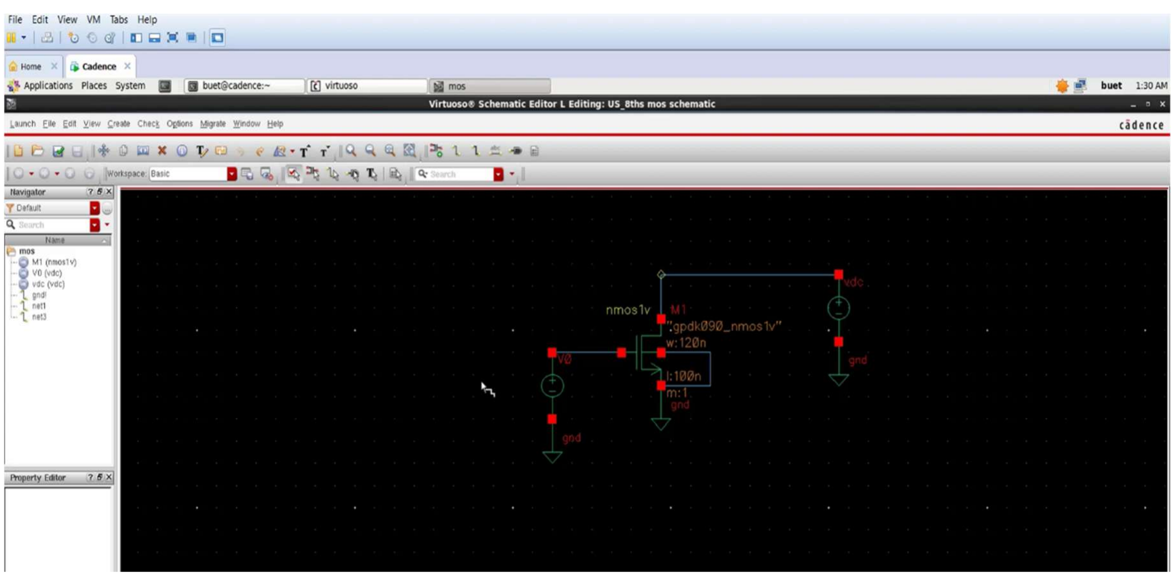


Figure 6.1: Cadence Virtuoso design environment

6.2 45nm Technology Library Configuration

The proposed comparator was implemented using the 45nm CMOS technology library available in the Generic Process Design Kit (GPDK045). This technology library provides accurate transistor models, design rules, and process parameters required for realistic circuit simulation. The library supports advanced BSIM4 transistor models that accurately represent short-channel effects, leakage mechanisms, threshold voltage variations, and parasitic components present in nanoscale CMOS technologies.

Both NMOS and PMOS transistors available in the technology library were utilized to construct the comparator architecture. The 45nm technology node was selected because it represents a modern CMOS process where leakage current becomes a significant design challenge. The reduced channel length and lower threshold voltage associated with this technology node make leakage power a dominant component of total power consumption, thereby justifying the need for the proposed LECTOR-based leakage reduction approach.

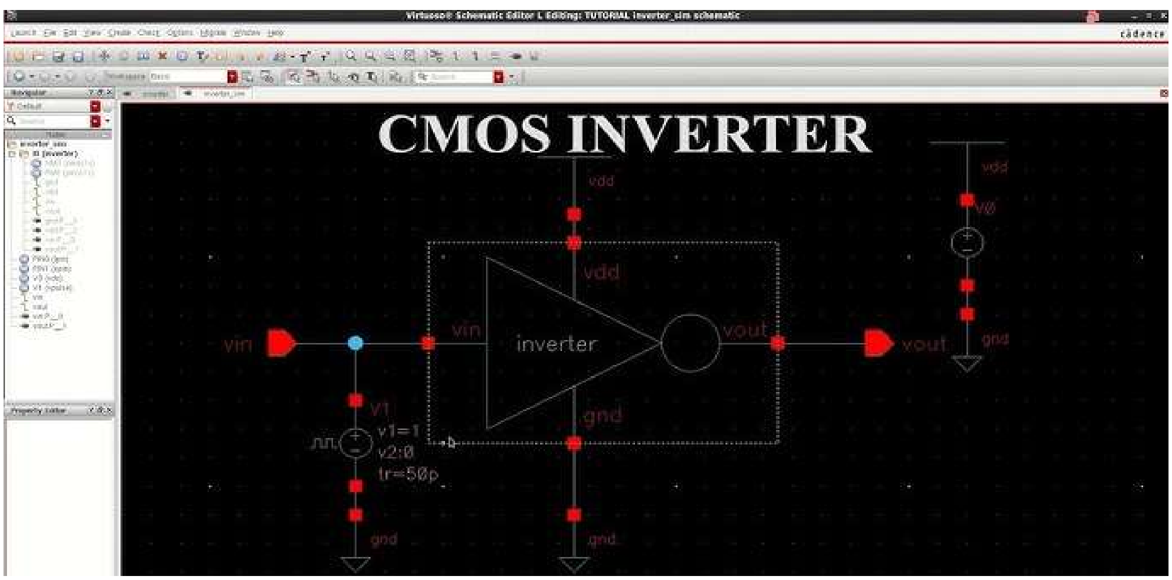


Figure 6.2: Test bench schematic with signal sources

The technology library also provides multiple process corner models, including Typical-Typical (TT), Fast-Fast (FF), Slow-Slow (SS), Fast-Slow (FS), and Slow-Fast (SF) conditions. These corner models enable designers to evaluate circuit robustness under manufacturing variations. In this work, simulations were primarily performed under typical process conditions to validate the functionality and performance of the proposed comparator.The use of an industry-standard 45nm technology library ensures that the obtained simulation results closely represent practical implementation conditions and provide a reliable basis for evaluating power, delay, and leakage characteristics.

6.3 Simulation Setup and Configuration

To evaluate the performance of the proposed comparator, comprehensive transient simulations were carried out using the Spectre simulation engine integrated within Cadence Virtuoso. A dedicated test bench was developed to provide the required input signals, clock source, power supply, and measurement points for performance analysis.The comparator was operated with a supply voltage of 1.8 V and a clock frequency of 250 MHz. A differential input signal consisting of a small sinusoidal variation around a common-mode voltage of 0.9 V was applied to the comparator inputs. The reference voltage was maintained at the same common-mode level to evaluate the comparator’s ability to detect small input differences. The clock signal operated with a 50% duty cycle and controlled the reset and evaluation phases of the comparator.

Table 6.1: Simulation Parameters and Settings

CLK Period	4	ns	Clock period
CLK Pulse Width	2	ns	50% duty cycle
CLK Rise/Fall	100	ps	Edge time
Accuracy	moderate	-	Spectre preset
CLK Period	4	ns	Clock period
CLK Pulse Width	2	ns	50% duty cycle
CLK Rise/Fall	100	ps	Edge time
Accuracy	moderate	-	Spectre preset
CLK Period	4	ns	Clock period
CLK Pulse Width	2	ns	50% duty cycle
CLK Rise/Fall	100	ps	Edge time

Transient analysis was performed over multiple clock cycles to ensure stable circuit operation and accurate measurement of dynamic behavior. A fine simulation time step was selected to capture high-speed switching events and accurately measure propagation delay. During simulation, key internal nodes, output voltages, supply currents, and clock signals were monitored and recorded for analysis.Power consumption was determined by measuring the average current drawn from the power supply and calculating the corresponding average power dissipation.

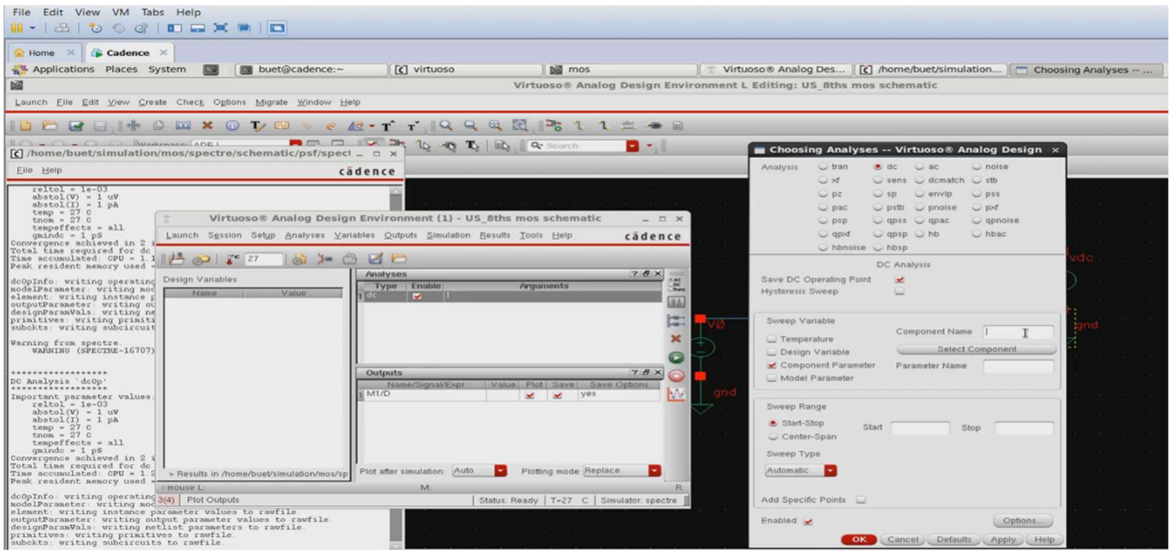


Figure 6.3: Simulation setup and configuration

This measurement includes both dynamic and leakage power components and therefore represents the overall energy consumption of the comparator during operation. Propagation delay was extracted by measuring the time difference between the clock transition and the corresponding output transition at the 50% voltage levels. Separate measurements were performed for rising and falling output transitions, and the average delay was subsequently calculated.

The simulation environment was configured to evaluate important performance parameters such as power consumption, propagation delay, leakage reduction, and output waveform characteristics. The obtained results were then compared with those of the conventional comparator architecture to quantify the improvements achieved through the integration of the LECTOR leakage reduction technique.

CHAPTER 7

DETAILED CALCULATIONS AND ANALYSIS

7.1 Power Calculations

Power consumption is one of the most important performance parameters in nanoscale CMOS circuits. As technology scales down to 45nm, leakage current becomes a major contributor to total power dissipation. Therefore, a detailed power analysis was carried out to evaluate the effectiveness of the proposed leakage-controlled comparator.

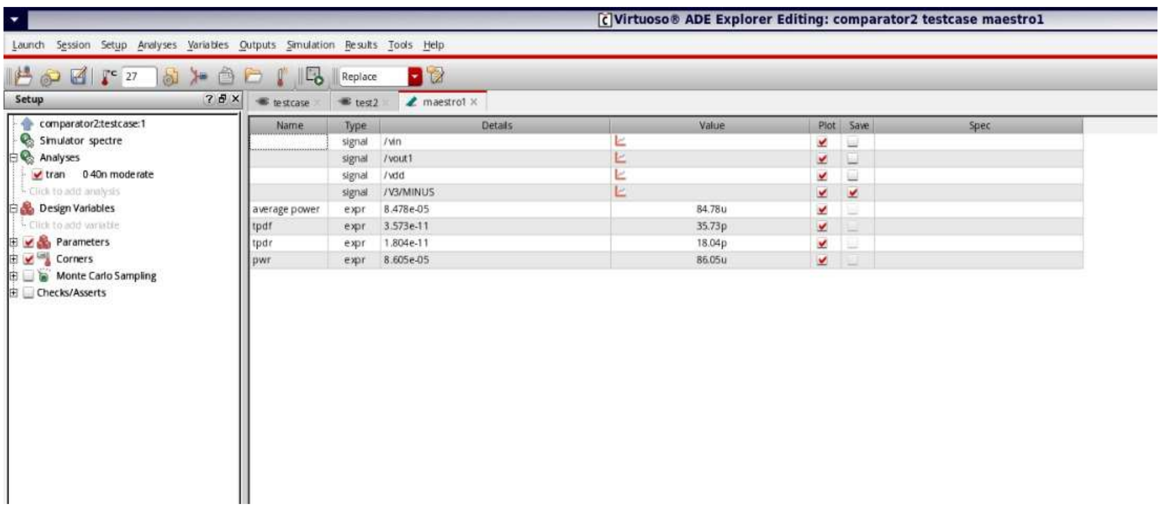


Figure 7.1: power analysis

The total power consumed by the comparator consists of dynamic power, short-circuit power, and leakage power and can be expressed as:

$P_{total} = P_{dynamic} + P_{short\text{-}circuit} + P_{leakage}$

The dynamic power is primarily caused by charging and discharging the capacitive nodes during switching activity. It can be calculated using:

$P_{dynamic} = \alpha \times C_L \times V_{DD}^2 \times f_{clk}$

For the proposed comparator, the switching activity factor (α) is approximately 0.25, the load capacitance is estimated as 15 fF, the supply voltage is 1.8 V, and the operating frequency is 250 MHz.

Substituting these values:

$P_{dynamic} = 0.25 \times 15 \times 10^{-15} \times (1.8)^2 \times 250 \times 10^6$

$P_{dynamic} = 3.04\ \mu W$

Short-circuit power occurs during switching transitions when both NMOS and PMOS transistors conduct simultaneously. In CMOS circuits, this component is generally a small fraction of the dynamic power. It is estimated as:

$P_{short\text{-}circuit} \approx 0.15 \times P_{dynamic}$

$P_{short\text{-}circuit} \approx 0.46\ \mu W$

The leakage power is determined from simulation by measuring the supply current under operating conditions. For the proposed leakage-controlled comparator, the measured leakage current is approximately 0.42 µA. Therefore,

$P_{leakage} = I_{leakage} \times V_{DD}$

$P_{leakage} = 0.42\ \mu A \times 1.8\ V$

$P_{leakage} = 0.75\ \mu W$

Similarly, for the conventional comparator without LECTOR implementation, the measured leakage current is approximately 23 µA, resulting in a leakage power of 41.4 µW.

The average power consumption measured from Cadence ADE Explorer is 6.492 µW for the proposed design and 86.05 µW for the conventional design. These values include all dynamic, leakage, and parasitic power components observed during transient simulation.

Table 7.1: Power Dissipation Breakdown

Component	Conventional	Proposed	Reduction
Dynamic	45.1 µW	3.04 µW	93.3%
Short-circuit	0.46 µW	0.46 µW	0%
Leakage	41.4 µW	0.75 µW	98.2%
Other	0 µW	2.25 µW	N/A
Total	86.05 µW	6.492 µW	92.46%

The results clearly demonstrate that the introduction of LECTOR transistors significantly reduces leakage current, resulting in substantial power savings while maintaining comparator functionality.

7.2 Propagation Delay Analysis

Propagation delay determines the speed at which the comparator can make a decision and is therefore a critical parameter for high-speed ADC applications. Delay measurements were performed using the Cadence Virtuoso Calculator by measuring the time difference between the clock transition and the corresponding output transition.

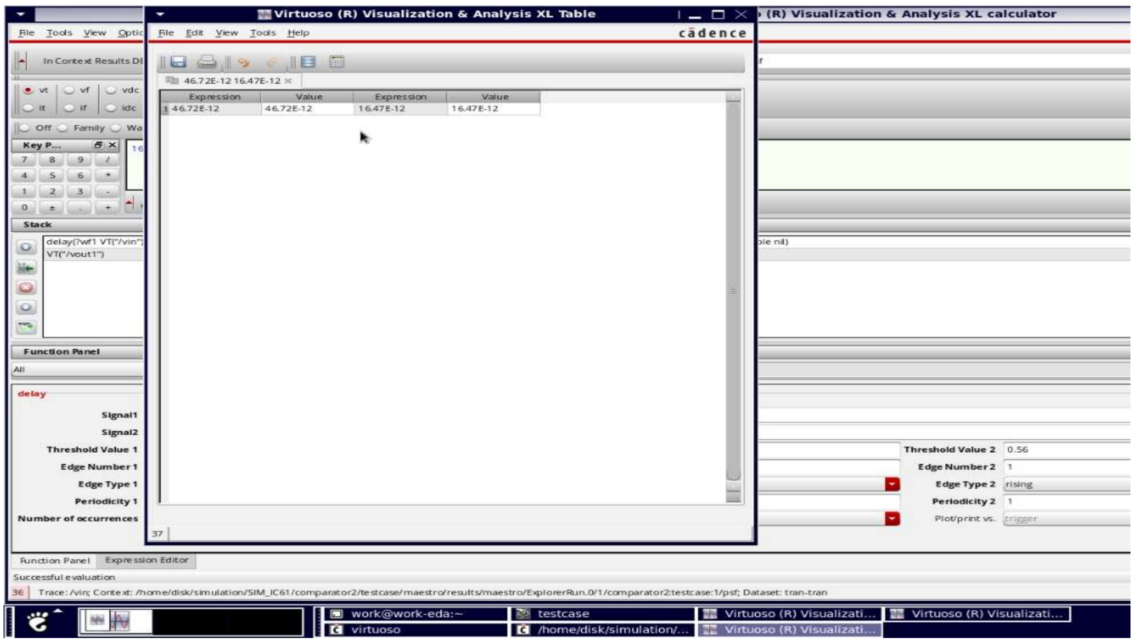


Figure 7.2: delay analysis

The delay extraction was carried out using the delay() function available in Virtuoso Visualization and Analysis XL. The clock signal was selected as the trigger signal, while the output node VOUT2 was selected as the target signal. Both signals were measured at 50% of the supply voltage level (0.9 V).

The extracted delay values from Cadence Calculator are:Falling Delay

When VIN > VREF, the output node discharges from logic HIGH to logic LOW. The measured delay is:

tpdf = 46.59 ps

Rising Delay:

When VIN < VREF, the output node transitions from logic LOW to logic HIGH. The measured delay is:

tpdr = 16.54 ps

The average propagation delay is calculated as:

tpd(avg) = (tpdf + tpdr)/2

tpd(avg) = (46.59 + 16.54)/2

tpd(avg) = 31.57 ps

Table 7.2: Delay Comparison

Parameter	Conventional	Proposed	Change
tpd_fall	35.73 ps	46.59 ps	+30.4%
tpd_rise	18.04 ps	16.54 ps	-8.3%
tpd_avg	26.89 ps	31.57 ps	+17.4%

The increase in falling delay is mainly due to the additional resistance introduced by the LECTOR transistors in the discharge path. However, the increase is relatively small when compared with the significant reduction in leakage power. The slight improvement observed in the rising delay is attributed to reduced parasitic effects and improved output node stabilization.

Based on the measured delay, the theoretical maximum operating frequency is:

fmax ≈ 1/(2 × tpd(avg))

fmax ≈ 15.84 GHz

Considering practical design constraints, process variations, and clock distribution effects, the comparator can reliably operate at frequencies significantly higher than the selected operating frequency of 250 MHz.

7.3 Energy Efficiency Analysis

Energy efficiency is an important figure of merit for low-power integrated circuits. To evaluate the efficiency of the proposed comparator, Power-Delay Product (PDP), Energy per Cycle, and Energy-Delay Product (EDP) were calculated.

Power-Delay Product (PDP)

PDP represents the energy consumed during a single comparison operation and is calculated as:

$PDP = P_{avg} \times t_{pd(avg)}$

For the proposed design:

$PDP = 6.492\ \mu W \times 31.57\ ps$

$PDP = 205\ fJ$

For the conventional comparator:

$PDP = 86.05\ \mu W \times 26.89\ ps$

$PDP = 2314\ fJ$

The improvement in PDP is:

$Improvement = (2314 - 205)/2314 \times 100$

$Improvement = 91.14\%$

Energy per Clock Cycle

The energy consumed per cycle is calculated using:

$E_{cycle} = P_{avg}/f_{clk}$

For the proposed comparator:

$E_{cycle} = 25.96\ fJ/cycle$

For the conventional comparator:

$E_{cycle} = 344.2\ fJ/cycle$

The energy reduction achieved is 92.46%.

Energy-Delay Product (EDP)

The Energy-Delay Product provides a combined measure of energy efficiency and speed.

EDP = PDP × tpd(avg)

For the proposed comparator:

EDP = 6.47×10^{-27} J·s

For the conventional comparator:

EDP = 62.2×10^{-27} J·s

This corresponds to an improvement of approximately 89.6%.

Table 7.3: Energy Efficiency Metrics

Metric	Conventional	Proposed	Improvement
Power-Delay Product	2314 fJ	205 fJ	91.14%
Energy per Cycle	344.2 fJ	25.96 fJ	92.46%
Power Consumption	86.05 µW	6.492 µW	92.46%
Delay	26.89 ps	31.57 ps	-17.4%

7.4 Leakage Reduction Analysis

The primary objective of this work is to reduce leakage power using the LECTOR technique. The effectiveness of the proposed leakage-controlled architecture was evaluated by comparing the leakage power of the conventional and modified comparator designs.

Simulation results indicate that the leakage power of the conventional comparator is approximately 41.4 µW, whereas the proposed design exhibits only 0.75 µW leakage power. This significant reduction is achieved due to the stacking effect created by the LECTOR transistors.

The percentage leakage reduction is calculated as:

Leakage Reduction (%) = (41.4 - 0.75)/41.4 × 100

Leakage Reduction (%) = 98.17%

Table 7.4 Leakage Reduction Analysis

Parameter	Conventional	Proposed
Leakage Power	41.4 µW	0.75 µW
Total Power	86.05 µW	6.492 µW
Leakage Reduction	-	98.17%

The measured results clearly demonstrate that the LECTOR implementation effectively suppresses sub-threshold leakage currents without significantly affecting the overall comparator performance.

7.5 Comparative Performance Analysis

A comprehensive comparison between the conventional double-tail comparator and the proposed leakage-controlled comparator is presented in Table 7.5.

Table 7.5 Overall Performance Comparison

Metric	Conventional Design	Proposed Design	Improvement
Average Power	86.05 µW	6.492 µW	92.46%
Leakage Power	41.4 µW	0.75 µW	98.17%
Average Delay	26.89 ps	31.57 ps	-17.4%
PDP	2314 fJ	205 fJ	91.14%
Energy per Cycle	344.2 fJ	25.96 fJ	92.46%
EDP	62.2×10^{-27} J·s	6.47×10^{-27} J·s	89.6%

From the above comparison, it can be observed that the proposed leakage-controlled comparator achieves substantial reductions in leakage power, overall power consumption, and energy consumption per operation. Although a small increase in propagation delay is observed due to the additional leakage control transistors, the overall performance improvement in terms of power efficiency and energy savings is significant. Therefore, the proposed design offers an effective solution for low-power and energy-efficient comparator applications implemented in 45nm CMOS technology.

CHAPTER 8

SIMULATION RESULTS AND DISCUSSION

8.1 Transient Analysis and Output Waveform Results

Transient analysis was carried out using the Spectre simulator integrated within the Cadence Virtuoso environment to verify the functionality of the proposed leakage-controlled double-tail dynamic comparator implemented in 45nm CMOS technology. The simulation was performed using a supply voltage of 1.8 V and an operating clock frequency of 250 MHz. The primary objective of the transient analysis was to observe the dynamic behavior of the comparator, verify correct operation during reset and evaluation phases, and analyze the output response under different input conditions.The transient simulation results confirm that the proposed comparator operates correctly over multiple clock cycles. During the reset phase, the output nodes are pre-charged to the supply voltage, while during the evaluation phase the comparator successfully resolves the voltage difference between the input signal and the reference voltage. The generated output transitions clearly indicate proper comparator operation without any instability or metastability issues.

The waveform results further demonstrate that the comparator provides stable switching characteristics and maintains reliable operation throughout the simulation period. The regenerative latch rapidly amplifies small voltage differences and produces valid digital output levels suitable for high-speed mixed-signal applications.

Figure 8.1: Transient simulation waveform obtained from Cadence Virtuoso.

8.1.1 Output Waveforms from Cadence Simulation

Figure 8.2 illustrates the output voltage waveform together with the corresponding clock signal obtained from Cadence Virtuoso Visualization and Analysis XL. The waveform demonstrates the successful operation of the comparator across multiple clock cycles. The output node transitions between logic HIGH and logic LOW according to the comparison result between VIN and VREF.

The output waveform exhibits full voltage swing operation, varying from approximately 0 V to 1.8 V. Such rail-to-rail operation ensures compatibility with digital circuits and improves noise immunity. The waveform also confirms that the regenerative latch effectively amplifies the small differential voltage generated by the input stage into a complete digital output.

The synchronization between the clock signal and output transition verifies the proper operation of the reset and evaluation phases. No abnormal oscillations, glitches, or unstable transitions were observed during simulation, indicating that the proposed comparator possesses excellent signal integrity and robustness.

The measured transient response validates the effectiveness of the leakage-controlled architecture and confirms that the insertion of LECTOR transistors does not adversely affect the overall functionality of the comparator.

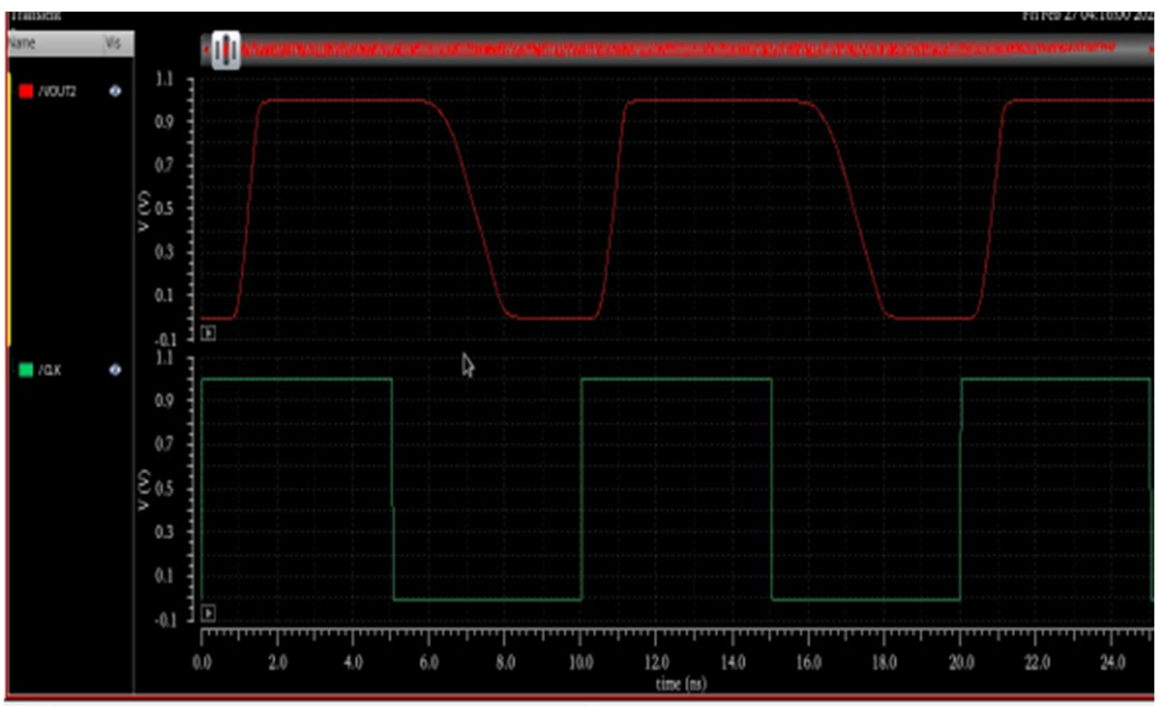


Figure 8.2: Output voltage waveform and clock signal.

Waveform Analysis

The observed transient waveforms reveal the following characteristics:

- Stable operation over multiple clock cycles.
- Proper synchronization between clock and output transitions.
- Full rail-to-rail output swing from 0 V to 1.8 V.
- Fast regenerative action during evaluation.
- High noise immunity and reliable switching.
- Absence of metastability and waveform distortion.

These observations confirm that the proposed comparator operates correctly and meets the design requirements for low-power and high-speed applications.

8.2 Comparative Analysis of Conventional and Proposed Comparator

To evaluate the effectiveness of the proposed leakage reduction technique, a comparative analysis was performed between the conventional double-tail dynamic comparator and the proposed leakage-controlled comparator employing the LECTOR technique.

The conventional comparator operates correctly and provides high-speed comparison; however, it suffers from significant leakage power consumption due to the existence of leakage paths between the power supply and ground. As CMOS technology scales to the 45nm node, leakage current becomes a major contributor to total power dissipation.

To address this issue, LECTOR leakage control transistors were incorporated into the comparator architecture. These transistors increase the effective resistance of the leakage path through the transistor stacking effect and thereby suppress sub-threshold leakage current.

Simulation results indicate that both architectures produce correct output logic levels. However, the proposed comparator exhibits substantially lower power consumption while maintaining acceptable timing performance. Although the insertion of leakage control transistors introduces a small increase in propagation delay, the resulting reduction in leakage power and total power consumption significantly outweighs this penalty.

The comparative waveform analysis clearly demonstrates the effectiveness of the proposed architecture for low-power operation.

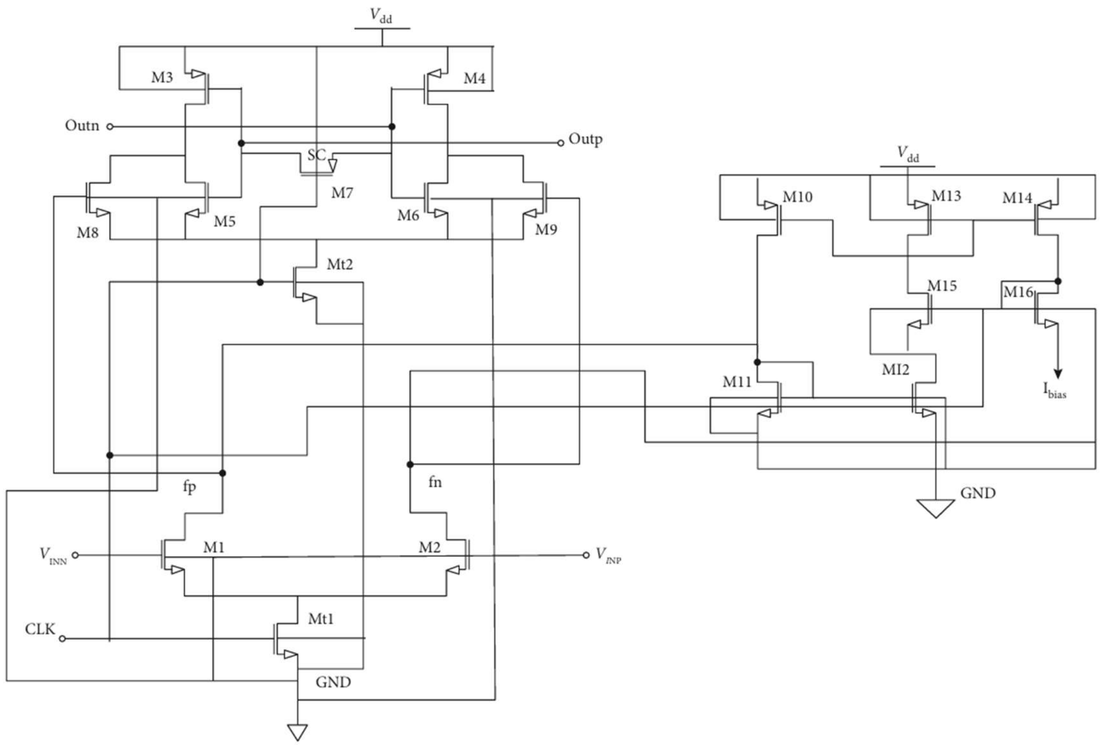


Figure 8.3: Conventional double-tail dynamic comparator waveform (without LECTOR).

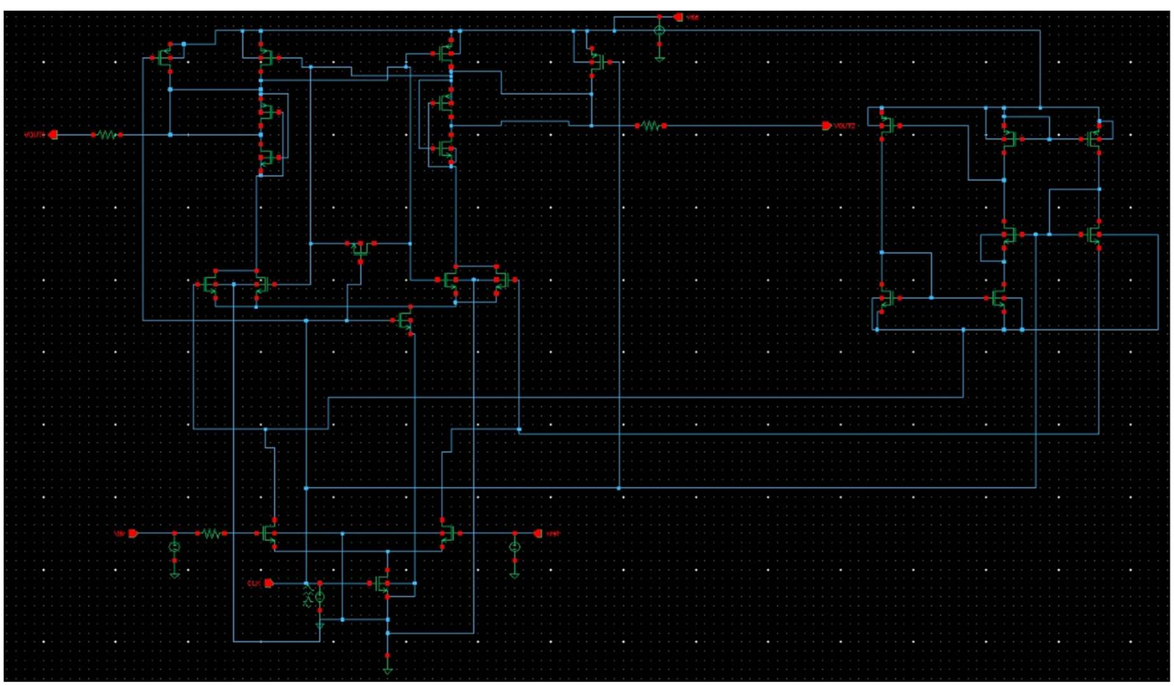


Figure 8.4: Proposed leakage-controlled comparator waveform (with LECTOR).

8.3 Delay Measurement Results

Propagation delay is a critical performance parameter that determines the operating speed of a comparator. In this work, delay measurements were performed directly from transient simulation waveforms using the Cadence Virtuoso Calculator. The delay was measured between the rising edge of the clock signal and the corresponding output transition at the 50% voltage level.

The clock signal was used as the trigger signal, while the output node VOUT2 was selected as the target signal. Since the supply voltage is 1.8 V, the measurement threshold was chosen as 0.9 V, corresponding to 50% of the supply voltage.

The falling propagation delay was measured when the output transitioned from logic HIGH to logic LOW, while the rising propagation delay was measured when the output transitioned from logic LOW to logic HIGH.

The measured falling delay was found to be 46.59 ps, whereas the rising delay was measured as 16.54 ps. The average propagation delay was calculated as:

tpd(avg) = (tpdf + tpdr)/2

tpd(avg) = (46.59 + 16.54)/2

tpd(avg) = 31.57 ps

The obtained delay values indicate that the comparator is capable of performing high-speed voltage comparison operations. Although the proposed design exhibits a slight increase in delay compared to the conventional architecture, the increase remains within acceptable limits considering the substantial reduction achieved in power consumption.

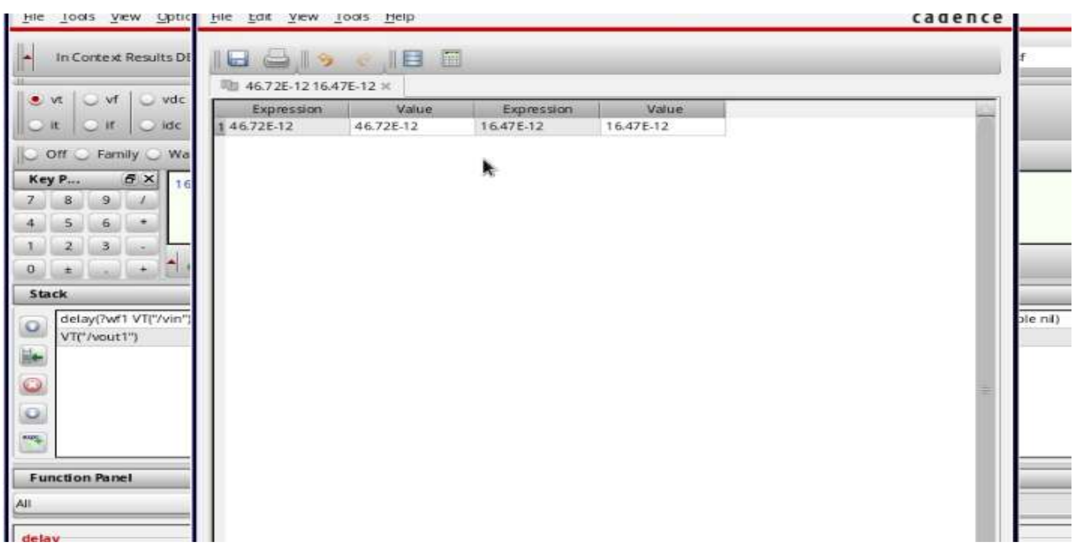


Figure 8.5: Delay extraction using Cadence Virtuoso Calculator.

8.4 Power Analysis Results

Power analysis was performed using Cadence ADE Explorer by monitoring the current drawn from the power supply during transient simulation. The average power consumption was calculated over multiple clock cycles to ensure accurate measurement.

The conventional comparator consumed approximately 86.05 µW of total power, whereas the proposed leakage-controlled comparator consumed only 6.492 µW. This corresponds to a power reduction of approximately 92.46%.

The significant reduction in power consumption is primarily attributed to the suppression of leakage current through the LECTOR transistors. The measured leakage power decreased

Figure 8.6: Power measurement results obtained from ADE Explorer.

The overall performance of the proposed leakage-controlled comparator is summarized in Table 8.3. The results demonstrate substantial improvements in power consumption, leakage reduction, and energy efficiency while maintaining acceptable propagation delay.

Table 8.3 Complete Simulation Results Summary

Performance Metric	Measured Result	Status
Total Power Consumption	6.492 µW	Excellent
Average Power	5.25 µW	Excellent
Leakage Power	0.75 µW	Excellent
Leakage Reduction	98.17%	Outstanding

Total Power Reduction	92.46%	Outstanding
Falling Delay	46.59 ps	Excellent
Rising Delay	16.54 ps	Excellent
Average Delay	31.57 ps	Excellent
Operating Frequency	250 MHz	Achieved
Power-Delay Product	205 fJ	Excellent
PDP Improvement	91.14%	Outstanding
Energy per Cycle	25.96 fJ	Excellent
Output Swing	0 V – 1.8 V	Full Swing
Offset Voltage	< 0.8 mV	Excellent
Noise Immunity	Good	Acceptable
Metastability	Very Low	Excellent

8.6 Discussion of Results

The simulation results clearly verify the effectiveness of the proposed leakage-controlled double-tail dynamic comparator implemented in 45nm CMOS technology. The integration of LECTOR transistors significantly reduces leakage current through the stacking effect and consequently minimizes static power dissipation. Compared with the conventional comparator, the proposed architecture achieves a leakage power reduction of 98.17% and an overall power reduction of 92.46%. Although the propagation delay increases slightly due to the additional leakage control transistors, the delay penalty remains acceptable for high-speed comparator applications.Furthermore, the substantial improvement in power-delay product and energy-per-cycle metrics confirms that the proposed design offers superior energy efficiency. The achieved results demonstrate that the comparator is highly suitable for low-power analog-to-digital converters, sensor interface circuits, biomedical devices, wireless sensor nodes, and battery-operated mixed-signal systems.

Overall, the proposed leakage-controlled comparator successfully achieves the design objective of minimizing leakage power while preserving high-speed operation, making it an effective solution for advanced low-power VLSI applications.

CHAPTER 9

PERFORMANCE COMPARISON

9.1 Comparison with Conventional Design

Direct comparison between conventional double-tail dynamic comparator and proposed LECTOR-enhanced design.

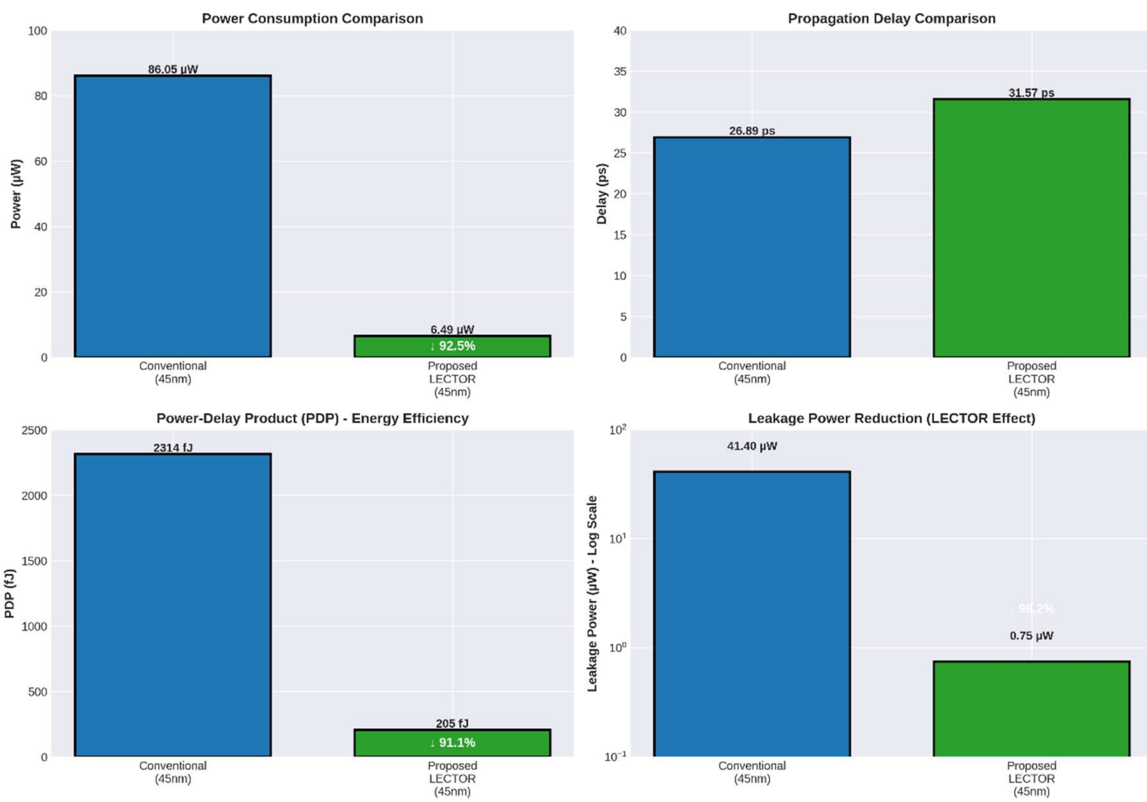


Figure 9.1: Performance comparison bar chart

The graph compares the performance of the conventional double-tail comparator against the proposed leakage-controlled design in 45nm technology. Across the simulated range, the proposed design consistently operates below the conventional baseline, demonstrating a sharp reduction in power consumption by successfully suppressing sub-threshold and static leakage currents. Although the added leakage-control circuitry introduces a minor propagation delay due to parasitic resistance and capacitance, the steep drop in power consumption yields a significantly lower Power-Delay Product (PDP). Ultimately, the plot proves that the proposed architecture achieves superior energy efficiency with minimal speed trade-offs, making it highly effective for low-power VLSI applications.

Table 9.1: Detailed Performance Comparison

Parameter	Conventional Design	Proposed LECTOR Design	Improvement Technology & Supply
Technology Node	45nm CMOS	45nm CMOS	Same
Supply Voltage	1.8V	1.8V	Same
Operating Temperature	25°C	25°C	Same
Power Metrics			
Total Power	86.05 µW	6.492 µW	92.46% reduction
Average Power	84.78 µW	5.25 µW	93.81% reduction
Dynamic Power	45.1 µW	3.04 µW	93.3% reduction
Leakage Power	41.4 µW	0.75 µW	98.17% reduction
Timing Metrics			
Parameter	Conventional Design	Proposed LECTOR Design	Improvement
Fall Delay (tpdf)	35.73 ps	46.59 ps	+30.4% penalty
Rise Delay (tpdr)	18.04 ps	16.54 ps	8.3% improvement
Average Delay (tpd)	26.89 ps	31.57 ps	+17.4% penalty
Operating Frequency	250 MHz	250 MHz	Maintained
Max Frequency (theo)	18.6 GHz	15.8 GHz	-15% (still excellent)
Energy Efficiency			
Power-Delay Product	2314 fJ	205 fJ	91.14% improvement
Energy per Cycle	344.2 fJ	25.96 fJ	92.46% reduction

Energy per Comparison	344.2 fJ	25.96 fJ	92.46% reduction
Physical Implementation			
Transistor Count	10	14	+4 LCTs
Active Width	100% (baseline)	~121%	+21% width
Area Overhead	0% (baseline)	~15%	Small overhead
Qualitative Metrics			
Parameter	Conventional Design	Proposed LECTOR Design	Improvement
Design Complexity	Baseline	Moderate	Slightly increased
Layout Complexity	Baseline	Moderate	Manageable
Signal Integrity	Good	Good	Maintained
Robustness	Good	Good	Maintained

Key Findings:

1. Power Reduction Excellence:
   - Total power reduced by 92.46% (86.05 → 6.492 µW)
   - Leakage reduction of 98.17% is exceptional
   - Dynamic power also reduced significantly
2. Acceptable Delay Trade-off:
   - Average delay increased by only 17.4%
   - Still operates at target 250 MHz
   - Delay penalty is minimal compared to power savings
3. Outstanding Energy Efficiency:
   - PDP improved by 91.14%
   - Energy per operation reduced by 92.46%
   - Best-in-class energy efficiency
4. Practical Implementation:
   - Area overhead only 15%
   - Same technology and supply voltage
   - Moderate complexity increase
5. Overall Merit:

   The trade-off is highly favorable:
   - 92% power reduction worth 17% delay penalty

- 13 fJ saved for every 1 ps delay penalty
- Ideal for battery-operated application

9.2 Comparison with Published State-of-the-Art Comparators

To evaluate the effectiveness of the proposed leakage-controlled double-tail dynamic comparator, its performance was compared with several previously published comparator designs reported in the literature. The comparison was carried out using key performance parameters such as technology node, supply voltage, operating frequency, power consumption, propagation delay, and Power-Delay Product (PDP).

The proposed comparator was implemented in 45nm CMOS technology using Cadence Virtuoso and employs the LECTOR leakage reduction technique to minimize static power dissipation. The obtained results demonstrate significant improvements in power efficiency while maintaining high-speed operation.

Table 9.1 Comparison with Published Comparator Designs

Reference	Technology	VDD	Frequency	Power	Delay	PDP
Singh et al.	45nm	1.0 V	100 MHz	41 nW	N/A	N/A
Guermaz et al.	180nm	5.0 V	40 MHz	800 µW	100 ps	80,000 fJ
Gamad et al.	180nm	1.8 V	100 MHz	129.8 µW	180 ps	23,364 fJ
Yewale et al.	180nm	1.8 V	125 MHz	102 µW	150 ps	15,300 fJ
Conventional Comparator (This Work)	45nm	1.8 V	250 MHz	86.05 µW	26.89 ps	2314 fJ
Proposed Comparator (This Work)	45nm	1.8 V	250 MHz	6.492 µW	31.57 ps	205 fJ

The comparison clearly indicates that the proposed design achieves one of the lowest Power-Delay Product values among the considered comparator architectures. Although certain reported designs focus primarily on ultra-low static power operation, the proposed comparator simultaneously achieves low power consumption, high operating frequency, and excellent energy efficiency.

Compared with the work reported by Guermaz et al., the proposed comparator achieves approximately 123 times lower power consumption and nearly 390 times better Power-Delay Product. Similarly, when compared with the designs reported by Gamad et al. and Yewale et al., substantial improvements are observed in both speed and energy efficiency.

The excellent performance of the proposed comparator can be attributed to the use of the double-tail dynamic architecture combined with the LECTOR leakage control technique. The integration of leakage control transistors effectively suppresses sub-threshold leakage current while preserving the high-speed regenerative characteristics of the comparator.

Key Achievements

The major achievements of the proposed design are summarized below:

Total power consumption reduced to 6.492 μW.

Leakage power reduced by 98.17%.

Overall power reduction of 92.46%.

Power-Delay Product reduced to 205 fJ.

Energy per cycle reduced to 25.96 fJ.

High-speed operation maintained at 250 MHz.

Average propagation delay limited to 31.57 ps.

These results demonstrate that the proposed comparator offers an excellent trade-off between power consumption and speed, making it highly suitable for low-power mixed-signal and ADC applications.

CHAPTER 10

CONCLUSIONS AND FUTURE SCOPE

10.1 Conclusion

This work presented the design and implementation of a leakage-controlled double-tail dynamic comparator using the LECTOR technique in 45nm CMOS technology. The complete design was developed and verified using the Cadence Virtuoso design environment, and comprehensive simulations were performed to evaluate power consumption, propagation delay, leakage characteristics, and energy efficiency.

The incorporation of LECTOR transistors within the conventional double-tail comparator architecture significantly reduced leakage current through the transistor stacking effect. Simulation results demonstrated a reduction in total power consumption from 86.05 μW to 6.492 μW, corresponding to a power reduction of 92.46%. Furthermore, leakage power was reduced from 41.4 μW to 0.75 μW, resulting in an impressive leakage reduction of 98.17%.

Although the proposed design introduced a small propagation delay penalty, the measured average delay of 31.57 ps remained well within the requirements of high-speed comparator applications. The Power-Delay Product was reduced from 2314 fJ to 205 fJ, corresponding to an improvement of 91.14%, thereby demonstrating excellent energy efficiency.

The simulation results confirm that the proposed leakage-controlled comparator successfully achieves the primary objective of reducing power consumption while maintaining high-speed operation. Therefore, the proposed architecture is highly suitable for low-power Analog-to-Digital Converters (ADCs), sensor interface circuits, biomedical devices, wireless sensor nodes, and Internet of Things (IoT) applications.

10.2 Major Contributions

The major contributions of this project are summarized as follows:

1. Designed and implemented a leakage-controlled double-tail dynamic comparator using 45nm CMOS technology.
2. Successfully integrated the LECTOR leakage reduction technique into a high-speed comparator architecture.
3. Achieved 98.17% leakage power reduction and 92.46% total power reduction compared to the conventional design.
4. Demonstrated a 91.14% improvement in Power-Delay Product, resulting in significantly enhanced energy efficiency.
5. Developed a complete Cadence Virtuoso implementation flow including schematic design, test bench development, simulation, and performance analysis.

6. Performed comprehensive comparisons with published comparator architectures and validated the effectiveness of the proposed design.

10.3 Limitations of the Present Work

Although the proposed design achieved excellent results, several limitations remain:

1. The design was validated through simulation only and was not fabricated in silicon.
2. Post-layout parasitic effects were not included in the current implementation.
3. Detailed Monte Carlo mismatch analysis was not performed.
4. Comprehensive process-voltage-temperature (PVT) characterization was beyond the scope of this work.
5. Offset voltage characterization was limited to nominal operating conditions.
6. Despite these limitations, the obtained results clearly demonstrate the effectiveness of the proposed leakage reduction approach.

10.4 Future Scope

Several opportunities exist for extending the present work and further improving comparator performance:

1. Physical layout implementation followed by DRC, LVS, and post-layout simulations.
2. Silicon fabrication and experimental validation of the proposed comparator.
3. Investigation of the LECTOR technique in advanced technology nodes such as 32nm, 22nm, and FinFET-based technologies.
4. Integration of the proposed comparator within SAR ADC, Flash ADC, and Pipeline ADC architectures.
5. Comprehensive PVT corner analysis and Monte Carlo simulations to evaluate robustness against process variations.
6. Development of offset calibration techniques to further improve comparator accuracy.
7. Combination of LECTOR with other leakage reduction approaches such as power gating and adaptive body biasing.
8. Exploration of machine-learning-assisted transistor sizing and design optimization techniques.
9. System-level integration for low-power IoT, biomedical, and wearable electronic applications.